Mean rational complexity in integer bases: a uniform bound and a two-prime theorem

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Abstract

For a uniformly chosen residue modulo B^N, let its rational height be the smallest $\max(|a|, d)$ among congruent rational representations $a \equiv sd \pmod{B^N}$, with $d > 0$ coprime to the entire modulus. We prove a uniform upper bound strictly below $N/2 + 0.241933621228105$ for the mean base-B logarithm of this height. The additive correction can be removed for prime-power bases, and is strictly negative for bases with exactly two distinct prime factors. The latter result combines explicit estimates with finite integer certificates covering arbitrary positive valuations. The correction cannot be removed for all bases: $B = 30, N = 2$ is a certified counterexample. An explicit lower estimate also gives an eventually positive family with radical 510510. The argument places the optimal global additive correction between 0.023672606561106 and 0.241933621228105. The sharper estimate combines supportwise positive and negative logarithmic contributions, with explicit finite premises, infinite tails and Kanold's classical unit-gap bound. Two simpler proofs give bounds 49/50 and 4/5 without the sharper-bound certificates. Its supremum reduces to finitely many bases, though neither its value nor finite-length attainment is determined here. Classical rational-reconstruction and lattice tools are distinguished from the new bounds. The companion specifies retained evidence, independent checks and full reconstruction commands.

1 Model, results and antecedents

Throughout, logarithms without a subscript are natural logarithms. All probability spaces in this paper contain *all* residues, not only invertible states.

Definition 1.1. For $M \geq 2$ and $s \in \mathbb{Z}/M\mathbb{Z}$, set

$$R_M(s) = \min_{\substack{d \in \mathbb{Z}_{>0},\ \gcd(d,M)=1 \\ a \in \mathbb{Z},\ a \equiv sd\ (M)}} \max(|a|, d).$$

For integers $B \geq 2, N \geq 1$, write $R_{B,N} = R_{B^N}$, with s uniformly distributed on $\mathbb{Z}/B^N\mathbb{Z}$, and put

$$E_{B,N} = \mathbb{E} \log_B R_{B,N}, \qquad \mu(M) = \mathbb{E} \log(R_M/\sqrt{M}).$$

Thus $E_{B,N} - N/2 = \mu(B^N)/\log B$.

The minimum exists: $d = 1$ with a centered representative of s gives $R_M(s) \leq \max(1, \lfloor M/2 \rfloor)$. In particular $R_M(0) = 1$. An admissible witness can be made primitive by dividing a, d by their gcd; the divisor is coprime to M. The full-unit condition on d is essential.

Theorem 1.2. *For every $N \geq 1$:*

1. *For every integer $B \geq 2$, $E_{B,N} < N/2 + 0.241933621228105$.*
2. *If B is a power of a prime, $E_{B,N} \leq N/2$.*

3. If B has exactly two distinct prime factors, $E_{B,N} < N/2$.

The binary case of the second assertion is the theorem of [1, Theorem 1.1]; its proof is imported and attributed here. The odd-prime argument below is independent of limiting laws. The two-prime theorem includes $B = p^a r^b$ with arbitrary $a, b \geq 1$; it is not restricted to squarefree bases or equal valuations.

Theorem 1.3. *One has $E_{30,2} > 1$. If $\operatorname{rad}(B) = q_0 = 510510$ and $B^N \geq q_0^2$, then*

$$E_{B,N} > N/2 + \frac{1}{1000 \log B}.$$

Consequently the zero-correction bound does not hold for all bases.

Theorem 1.4. *Set*

$$c(B) = \sup_{N \geq 1}(E_{B,N} - N/2), \quad c_+(B) = \max(0, c(B)), \quad C^* = \sup_{B \geq 2} c(B).$$

Then $c_+(B) \to 0$ as $B \to \infty$, and $0.023672606561106 < C^* < 0.241933621228105$. There exists a finite B_0 such that $C^* = \max_{2 \leq B \leq B_0} c(B)$. This does not assert that the supremum over N is attained.

Question 3.2 of the survey [2, p. 20282] asks for an upper bound on the expected finite k-adic complexity for k-ary strings. The following comparison fixes the rational-complexity convention used in our answer.

Equality of the rational observables. Let $S^N = (s_0, \ldots, s_{N-1})$ be a B-ary string, and let $\lambda_B(S^N, N)$ be the minimum unrounded $\log_B \max(|a|, |d|)$ over reduced rational numbers a/d whose B-adic expansions begin with S^N. Set $s = \sum_{j=0}^{N-1} s_j B^j \pmod{B^N}$. Then

$$\lambda_B(S^N, N) = \log_B R_{B^N}(s).$$

Indeed, a reduced rational has a B-adic expansion precisely when its denominator is coprime to B. Agreement of the first N digits is equivalent to $a \equiv sd \pmod{B^N}$. Conversely every full-unit congruent pair gives such an expansion. Eventual periodicity follows from the digit recursion $a_{j+1} = (a_j - d s_j)/B$: its integer numerators eventually lie in a bounded interval, so the deterministic recursion repeats. Cancelling common content preserves the congruence because the content is a unit modulo B^N, and changing both signs makes $d > 0$, without increasing the height. The string-to-residue map is a bijection. Thus the uniform expected rational complexity is exactly $E_{B,N}$, and Theorem 1.2(1) answers Question 3.2 in this convention. This agrees with the rational minimum in the author draft [3, Section 21.5].

A connection integer in an FCSR realization may have the additional condition $d \equiv -1 \pmod{B}$, and register lengths may be rounded. The equality above does not assert equivalence with those constrained or rounded quantities, which require a separate comparison. The cited author draft of the monograph [3, Sections 19.3,21.3,21.5] treats coprime rational reconstruction and complexity estimates; its Section 19.1 uses a different connection-integer convention. Section locators here refer to that author draft, not a verified pagination of the published edition. The known lower estimates already give the leading coefficient $1/2$ for fixed bases. We do not claim that coefficient as a new result.

Chen–Winterhof [4] gives probabilistic binary results; its accessible preprint version has been checked, while the final published text has not been fully accessed for the source comparison reported here. The exact binary mean and variance results in [1] are prior results, not new claims here. Farey fractions with excluded denominators were studied by Boca–Heersink–Spiegelhalter [6]; congruence sections by Heersink [7]; smallest-denominator laws and logarithmic moments by Marklof [8, 9]. Those results are relevant antecedents, but an unmarked law cannot simply be rescaled to impose all the full-unit restrictions. Priority of the present bounds is not certified by the source comparison presented here.

2 A lattice upper bound

For a squarefree integer q, let $J(q)$ be the maximum gap between successive integers coprime to q, including the cyclic gap modulo q. Let λ_s be the shortest Euclidean length of a nonzero vector in

$$\Lambda_s = \{(a,d) \in \mathbb{Z}^2 : a \equiv sd \pmod{M}\}, \qquad \det \Lambda_s = M.$$

Lemma 2.1. *For $q = \mathrm{rad}(M)$,*

$$R_M(s) \le \sqrt{1 + J(q)^2/3}\,\frac{M}{\lambda_s}.$$

Also $J(q) \le 2 \cdot 4^{\omega(q)}$, uniformly over squarefree q.

Proof. A shortest vector u is primitive in the lattice and extends to a basis u, v. After subtracting an integer multiple of u, write $v = \theta u + v_\perp$, where $|\theta| \le 1/2$ and $\|v_\perp\| = M/\lambda_s$. Minimality of u gives $\lambda_s^4 \le 4M^2/3$. The second coordinates of u, v have gcd one, since the lattice contains $(s, 1)$. If a prime dividing q divides the second coordinate of u, it cannot divide that of $v + ju$. For every other prime, precisely one residue class of j is forbidden. By CRT this is a translate of the unit pattern modulo a divisor of q. Its maximum gap is at most $J(q)$. Choose an allowed integer j within $J(q)/2$ of $-\theta$. Then $v + ju$ has a full-unit second coordinate and

$$\|v + ju\|^2 \le M^2/\lambda_s^2 + (J(q)^2/4)\lambda_s^2 \le (1 + J(q)^2/3)M^2/\lambda_s^2.$$

Changing the sign makes the denominator positive. If the denominator of u is already a unit, u itself is available; the same bound holds because $\lambda_s^2 \le 2M/\sqrt{3}$ and $J(q) \ge 1$.

For the second assertion, inclusion–exclusion shows that an interval of L consecutive integers contains at least $L\varphi(q)/q - 2^{\omega(q)}$ units. Since $\varphi(q)/q \ge 2^{-\omega(q)}$, $L = 2 \cdot 4^{\omega(q)}$ makes this quantity positive. Every such interval contains a unit. □

Lemma 2.2. *For $1 \le Y < M$, $\mathbb{P}(\lambda_s \le Y) \le 3Y^2/M$. Consequently, with $Z_M = R_M/\sqrt{M}$,*

$$\mathbb{P}(Z_M > z) \le \min(1, (J(q)^2 + 3)z^{-2}), \qquad \mu(M) \le \frac{\log(J(q)^2 + 3) + 1}{2}. \tag{2.1}$$

Proof. A horizontal nonzero lattice vector has length at least M. For $1 \le d \le Y$, put $g = \gcd(d, M)$. An integer a is incident with g residues s if $g \mid a$, and with none otherwise. Hence the number of incidences in the disk is at most

$$\sum_{1 \le d \le Y} (2\sqrt{Y^2 - d^2} + \gcd(d, M)) \le (1 + \pi/2)Y^2 < 3Y^2.$$

Here $\gcd(d, M) \le d$, $\sum_{d \le Y} d \le Y^2$ for $Y \ge 1$, and the decreasing semicircle sum is bounded by its integral. The union bound and Lemma 2.1 give the tail bound. When its threshold is below one there is no nonzero integral vector; when it is at least M, the stated probability bound is trivial. Finally

$$\mathbb{E} \log Z_M \le \mathbb{E}(\log Z_M)_+ \le \int_0^\infty \min(1, Ke^{-2t})\,dt = (\log K + 1)/2, \quad K = J(q)^2 + 3.$$

□

For two primes a slightly sharper tail is available:

$$\mathbb{P}(Z_M > z) \le \min(1, 6z^{-2}). \tag{2.2}$$

Indeed, if neither denominator of u, v is a full unit, the two bad-prime sets must be complementary. Then both $u+v$ and $u-v$ have unit denominators. The smaller Euclidean norm is at most $\sqrt{\|u\|^2+\|v\|^2-2|\langle u,v\rangle|} \le \sqrt{2}M/\lambda_s$ by the reduced-basis bounds. If one basis vector is a unit the same inequality follows directly. Combining with the disk count proves (2.2).

A first uniform bound. Prime-power bases satisfy the stronger second assertion, proved below. Suppose q has at least two primes and $q \ge 10$. If q is odd, the units $1, (q-1)/2, (q+1)/2, q-1$ show $J(q) \le q/2$. If $q = 2d$ is even and $d \ge 5$ is odd, the units $1, d-2, d+2, 2d-1$ give the same conclusion. Thus

$$J(q)^2 + 3 \le 7q^2/25, \qquad e(J(q)^2+3) < q^2,$$

using $e < 3$. Equation (2.1) gives $\mu(B^N) < \log q \le \log B$. For $q = 6$, $J(6) = 4$; if $B \ge 12$, $19e < 57 < 144 \le B^2$. For $B = 6$, (2.2) gives $\mu(M) \le (\log 6 + 1)/2 < \log 6$. All constants are independent of N. □

Proposition 2.3 (An explicit common correction)**.** *For all integer bases $B \ge 2$,*

$$c(B) \le \max(0, w, 39/40) < 49/50, \qquad w = 1 + \frac{\log(209/300)}{24\log 2}.$$

In particular $C^* < 49/50$. This is the elementary predecessor of Theorem 1.2(1); its stronger bound is proved in Section 8.

Proof. The prime-power and two-prime arguments of Sections 4 and 5 do not use this proposition; they give $c(B) \le 0$ when $k = \omega(q) \le 2$, where $q = \mathrm{rad}(B)$. For the other bases, (2.1) gives

$$c(B) \le \frac{\log(e(J(q)^2+3))}{2\log q}.$$

If $3 \le k \le 5$, then $q \ge 30$ and $J(q) \le \min(q/2, 2048)$. For $q \le 4096$, using $e < 11/4$,

$$\frac{e(J(q)^2+3)}{q^2} < \frac{11}{4}\left(\frac{1}{4} + \frac{3}{30^2}\right) = \frac{209}{300}.$$

Put $\delta = -\log(209/300) > 0$. Since $2\log q \le 24\log 2$, the common upper bound is $c(B) \le w$. For $q \ge 4096$, the fixed numerator $\log(e(2048^2+3))$ and the increasing denominator give the same bound, by evaluating at $q = 4096$. Moreover

$$\delta > \frac{91}{300} + \frac{1}{2}\left(\frac{91}{300}\right)^2 = \frac{62881}{180000} > \frac{42}{125}, \qquad 24\log 2 < \frac{84}{5}.$$

Thus $\delta/(24\log 2) > 1/50$, and $w < 49/50$.

For $k \ge 6$, the product of the first six distinct primes is at least 30030, and every further prime is at least $17 > 16$. Hence $q \ge 30030\,16^{k-6}$, while $J(q)^2 + 3 \le 5\,16^k$. The elementary enclosures $\log 30030 > 10$, $2/3 < \log 2 < 7/10$, $\log 5 < 17/10$ give

$$c(B) < \frac{(14/5)k + 27/10}{20 + (16/3)(k-6)} \le \frac{39}{40} < \frac{49}{50}.$$

The rational function equals $39/40$ at $k = 6$ and decreases; its derivative numerator is -48. For the only additional constant, $e < 11/4$ and $11^{10} < 30030\,4^{10}$ imply $\log 30030 > 10$. Taking the supremum over bases uses the two *common* caps w and $39/40$, rather than merely a pointwise strict inequality. No optimum is identified. □

3 Exact central counts and certified logarithms

Let $U_m(T) = \#\{1 \le d \le T : \gcd(d, m) = 1\}$. For $T^2 \le M$ and a proper divisor $g \mid M$, put $m = M/g$ and

$$P_g(T) = \sum_{\substack{e \le T/g \\ \gcd(e,M)=1}} \mu_{\mathrm{Mob}}(e)\, U_m\Big(\Big\lfloor \frac{T}{ge} \Big\rfloor\Big)\, U_q\Big(\Big\lfloor \frac{T}{e} \Big\rfloor\Big), \quad q = \mathrm{rad}(M). \tag{3.1}$$

Here μ_{Mob} denotes the Möbius function. These count primitive positive witnesses $a = ga'$, with $\gcd(a', m) = 1$ and full-unit denominator. Within either sign no two such primitive fractions coincide modulo M: their determinant has absolute value at most $T^2 - 1 < M$. Opposite signs can collide only when $af + ed = M$; the possible value $2M$ at $T^2 = M$ would force all four components equal to T, which cannot give a unit denominator. Write $J_g(T)$ for these collisions in the gcd-g state layer. Then

$$C_g(T) = 2P_g(T) - J_g(T), \qquad C(T) = 1 + \sum_{g \mid M,\ g < M} C_g(T). \tag{3.2}$$

The one is the zero state and must not be absorbed into a nonzero layer.

For completeness the collisions are finite exact computations: enumerate ordered full-unit $d, f \le T$ with $\gcd(d, f) = 1$, set $a_0 = Mf^{-1} \bmod d$ with $0 \le a_0 < d$, and $e_0 = (M - fa_0)/d$. All solutions are $a = a_0 + kd,\ e = e_0 - kf$. Intersect the two closed integer ranges $1 \le a, e \le T$; assign surviving pairs to $\gcd(a, M)$, enforcing primitivity. This handles endpoints, signs and empty phases exactly.

3.1 Lower CDFs outside the central range

For $M = p^a r^b$, $p < r$, a nonzero state in layer g is $s = gt$, where t is a unit modulo $m = M/g$. For $m > 2$, choose the centered inverse magnitude h and its sign $\varepsilon \in \{1, -1\}$, so that $1 \le h \le H = \lfloor (m-1)/2 \rfloor$ and $th \equiv \varepsilon \pmod m$. The witness $(\varepsilon g, h)$ is admissible if h is a full unit. If m is a pure power of one prime u and the missing prime v divides h, replacing h by $m - h$ repairs the denominator and changes the numerator sign. Put

$$V(x) = \lfloor x/v \rfloor - \lfloor x/(pr) \rfloor .$$

For $T \ge g, m > 2$ the resulting lower CDF in that layer is

$$W_g(T) = 2U_q(\min(T, H)) + 2\mathbf{1}_{\{m \text{ pure}\}} \max(0, V(H) - V(\max(0, m - T - 1))). \tag{3.3}$$

For $T < g$ it is zero. When $m = 2$ the layer contains one state, with witness height g. There is also the direct-denominator-one bound

$$B_g(T) = 2U_m(\min(\lfloor T/g \rfloor, H)) \tag{3.4}$$

for $T \ge g, m > 2$, with the same exceptional conventions. Each layer has the deterministic height cap

$$K_g = \min(M/p,\ g\lfloor m/2 \rfloor,\ \max(g, D_m)), \tag{3.5}$$

where $D_m = 1$ for $m = 2$, $D_m = H$ when m has both primes, and $D_m = \max(H, m - v)$ when m is pure. The first cap follows from the circle lemma below; the others follow from the displayed witnesses. Bounds from the same layer are combined by a maximum, not a sum.

Lemma 3.1 (Circle cap)**.** *For M with smallest prime divisor p, $R_M(s) \le \max(M/p, p - 1)$. If $M/p \ge p - 1$, the bound is M/p, and equality occurs for $s = M/p$. This condition holds for every two-prime modulus and every prime power of exponent at least two.*

Proof. Place $0, s, \ldots, (p-1)s$ on the circle of circumference M. If two coincide their index difference supplies numerator zero and a denominator between one and $p-1$. Otherwise their smallest cyclic gap is at most M/p, again with index difference between one and $p-1$. That denominator is coprime to M, proving the maximum bound. For $s = M/p$, every unit denominator is nonzero modulo p; its centered product with s has absolute value at least M/p. □

For $T^2 > 2M$, delete nonunit denominators from the Farey sequence of order T. Adjacent deleted denominators with complementary bad primes have unit difference; taking that difference repairs one neighbor. A gap can remain uncovered only next to a bad center $d < M/T$. Its two retained neighbors have denominators $> T - jd$, where $j = 1$ if $pr \mid d$, and $j = 2$ otherwise. These are positive because $T^2 > 2M$. Its open length in residue coordinates is at most $2(M - Td)/(d(T - jd))$. One extra integer per open hole gives

$$S(T) \le \sum_{\substack{d \le \lfloor (M-1)/T \rfloor \\ \gcd(d,M)>1}} \varphi(d) + \sum_{\substack{d \le \lfloor (M-1)/T \rfloor \\ \gcd(d,M)>1}} \frac{2\varphi(d)(M - Td)}{d(T - jd)}. \tag{3.6}$$

This is an upper survival bound, not an equality. The strict start $\left\lfloor \sqrt{2M} \right\rfloor + 1$ avoids the removable equality endpoint. The same Farey repair and length argument underlies the analytic estimates in the appendix.

3.2 Pointwise histograms for the initial finite bridges

For a two-prime modulus M, the base-six diagonal, sixth-power and squarefree-prefix batches use a pointwise construction, distinct from the aggregate CDF calendar. Put

$$k = \left\lfloor \sqrt{M} \right\rfloor, \qquad \tau = \left\lfloor \sqrt{2M} \right\rfloor + 1, \qquad D_1(s) = \max(1, \min(s, M - s)).$$

The closed central enumeration gives the exact height of every state with $R_M(s) \le k$, including zero separately. Apply the proved far-cover construction at τ, retaining its actual full-unit neighbors and phases, and let $\mathcal{H}$ be the union of the open exceptional holes. Outside $\mathcal{H}$, set $U(s) = \min(\tau, D_1(s))$, since the cover proves $R_M(s) \le \tau$. Inside $\mathcal{H}$, start with $D_1(s)$ and lower $U(s)$ only by heights $\max(|a|, d)$ of explicitly checked pairs

$$d > 0, \qquad \gcd(d, M) = 1, \qquad a \equiv sd \pmod{M}.$$

Every such pair is an admissible upper witness; overlaps are handled by minima. Restore the exact central heights wherever available. Consequently $R_M(s) \le U(s)$ pointwise, and its histogram has mass M.

For example, at a reduced exceptional center c/v, put $\Delta = Mc - vs$ and $\varepsilon = \operatorname{sign}(\Delta)$, with $\varepsilon = 1$ if $\Delta = 0$. For any positive full-unit d with $cd \equiv \varepsilon \pmod{v}$, the integer

$$a = \varepsilon(M - |\Delta| d)/v$$

satisfies $a - sd = M(\varepsilon - cd)/v$. Thus it is a legal witness; its optimality is neither assumed nor needed. The implementation checks integrality, congruence and denominator marks.

For $M = pr$, $p < r$, cap U by r, using Lemma 3.1. When $r \le \tau$, that cap suffices off the exact central set. For the actual histograms in these batches, the separately checked integer sign premise $Q < 0$, together with $R_M \le U$ and the outward logarithm enclosures below, proves the required negative centered mean. Pointwise domination alone does not imply that sign. For the diagonal moduli $M = 6^n$, $2M$ is nonsquare, so the historical threshold $\left\lceil \sqrt{2M} \right\rceil$ equals the strict threshold τ.

3.3 Integer logarithmic certificates

A calendar contains all integers up to an initial k, then steps by $T \mapsto \lceil 257T/256 \rceil$, with every layer cap, the far-tail start, and the terminal M/p inserted. At each calendar point combine (3.2) when applicable, the layer witness bounds, and (3.6); take cumulative maxima. Filling the gaps by the previous CDF value produces a lower CDF $\underline{C} \le C$. Its increments n_h give a dominating height histogram:

$$\sum_h n_h = M, \qquad \mathbb{E} \log R_M \le M^{-1} \sum_h n_h \log h.$$

The increments are not claimed to be the true height distribution.

Use outward integer ticks $L_S(x) \le S \log x \le U_S(x)$ and retain

$$Q = 2 \sum_h n_h U_S(h) - M L_S(M). \tag{3.7}$$

Then $Q < 0$ proves $\mu(M) < 0$, including odd M. The identity

$$\log x = 2 \sum_{j=0}^{K-1} \frac{z^{2j+1}}{2j+1} + \varepsilon, \quad z = (x-1)/(x+1), \quad 0 \le \varepsilon \le \frac{2z^{2K+1}}{(2K+1)(1-z^2)}$$

for rational $x \ge 1$, with range reduction by powers of two, supplies exact rational enclosures. No floating logarithm or assertion-sensitive test is used for the sign. The independent checker uses a different series depth and tick scale. It independently reconstructs the initial boundary, terminal boundary and relevant boundary gcd strata. It consumes the intervening central arrays; the central lemma and producer are therefore essential parts of the proof. It is not a complete independent recount of every intermediate CDF value.

4 Prime powers

For odd $M = p^n$ let $a = \sqrt{M}$, $K = \left\lfloor a/\sqrt{2} \right\rfloor$. The central count is

$$C(T) = 1 + 2P_p(T), \qquad P_p(T) = \#\{u, d \le T : \gcd(u,d) = 1,\ p \nmid d\}, \quad T \le K.$$

Indeed $2T^2 < M$, so opposite primitive witnesses do not collide. Before primitivity there are at least $(1-1/p)T^2$ pairs. The divisor union bound gives

$$P_p(T) \ge (1-y)\{2 - (1-y^2)\zeta(2)\}T^2 - (1-y)T \log T, \qquad y = 1/p.$$

To see the error explicitly, a common divisor $d \ge 2$ must satisfy $p \nmid d$, and $\lfloor x \rfloor (\lfloor x \rfloor - \lfloor x/p \rfloor) \le (1-y)(x^2 + x)$. Sum over d, use $\sum_{d \ge 2, p \nmid d} d^{-2} = (1-y^2)\zeta(2) - 1$ and $\sum_{d=2}^{T} d^{-1} \le \log T$. The elementary enclosure $\zeta(2) < 5/3$ follows by summing five terms and bounding the rest by an integral. Consequently

$$C(T) \ge cT^2 - dT \log T, \quad (c,d) = \begin{cases} (56/81, 4/3), & p = 3, \\ (5/8, 2), & p \ge 5. \end{cases} \tag{4.1}$$

The exact compensation identity is

$$\mu(M) = -D + P_+, \quad D = M^{-1} \int_1^a C(\lfloor x \rfloor) \frac{dx}{x}, \quad P_+ = M^{-1} \int_a^\infty S(\lfloor x \rfloor) \frac{dx}{x}. \tag{4.2}$$

Retaining the plateau $C(K)$ up to a yields

$$D \ge c(K^2/a^2)(1/2 + \log(a/K)) - 2cK/a^2 - d(K \log K/a^2)(1 + \log(a/K)). \tag{4.3}$$

For $a \ge 140$, $K/a \ge 11/16$, $K/a < 3/4$, $K^2/a^2 \ge 1/2 - \sqrt{2}/a$, and $\log 2/2 \le \log(a/K) \le 5/11$. Using $2/3 \le \log 2 \le 1$ and $\sqrt{2} < 3/2$ gives

$$D \ge \begin{cases} 70/243 - [56/27 + (16/11)\log a]/a, & p = 3, \\ 25/96 - [15/8 + (24/11)\log a]/a, & p \ge 5. \end{cases} \tag{4.4}$$

For integer $T \ge a$, the holes next to reduced Farey centers with $p \mid d, d < M/T$ have retained neighbors $b_L, b_R > T - d$. Their open lengths are at most $2(M - dT)/(d(T - d))$. Let $\kappa = (p-1)/p^2$, $t = T/a$, and

$$F(t) = \sum_{j \ge 1} \frac{t^{-2j}}{j(j+1)} = 1 + (t^2 - 1)\log(1 - t^{-2}), \quad F(1) = 1.$$

Summation of lengths and one integer per hole, using $\varphi(pk) \le (p-1)k$, gives

$$S(T)/M \le 2\kappa F(t) + \frac{\kappa a^2}{2T^2} + \frac{1 - 1/p}{2T}. \tag{4.5}$$

One can check the rounding term directly: the hole count is bounded by $(p-1)L(L+1)/2$, $L = \lfloor M/(pT) \rfloor$. The decreasing length summand is integrated with spacing p. Since F decreases and $\int_1^\infty F(t)\,dt/t = (\zeta(2) - 1)/2$, integrating the initial interval $[a, \lceil a \rceil]$ and retaining the first summand gives

$$P_+ \le \begin{cases} 11/54 + 17/(9a) + 1/(3a^2), & p = 3, \\ 11/75 + 19/(10a) + 1/(2a^2), & p \ge 5. \end{cases}$$

Thus

$$\mu(M) \le \begin{cases} -41/486 + [107/27 + (16/11)\log a]/a + 1/(3a^2), & p = 3, \\ -91/800 + [151/40 + (24/11)\log a]/a + 1/(2a^2), & p \ge 5. \end{cases} \tag{4.6}$$

At $a = 140$, using $\log 140 < 5$, both upper bounds are negative: the positive margins are at least $214289/52390800$ and $3811/431200$, respectively. All error terms decrease thereafter.

For $M = p \ge 5$, choose $k = \lfloor \sqrt{p} \rfloor$. Among the $(k+1)^2 > p$ pairs $0 \le u, v \le k$, two have the same $u - sv$; their difference has nonzero denominator at most $k < p$. Thus $R_p \le k < \sqrt{p}$. For $M = p^2$, the circle argument gives $R_M \le p = \sqrt{M}$. Only the 21 entries in Table 1 remain below 19600. Here $H = \sum_s R_M(s)$ and the exact inequality $H^2 < M^3$ implies $\mu(M) < 0$ by Jensen. The companion computes these minima without an assumed mean bound.

Prime	Exponents n	Sums H, in order
3	1–8	3,17,105,577,3107,16527,86977,455939
5	3–6	1009,11729,134011,1513793
7	3–5	4447,84975,1595511
11	3–4	32803,1220493
13	3	68697
17	3	225957
19	3	370511
23	3	866951

Table 1: The finite odd-prime bridge.

This proves Theorem 1.2(2) for odd prime powers. If $B = p^b$, replace n by bN; no new finite cases occur. For $p = 2$, use the exact theorem of [1].

5 The two-prime theorem

The appendix gives the full parametric compensation inequality and its refinements. Its proof uses the exact central formula, explicit errors for the unit-counting function, and the Farey survival estimate (3.6). It is an inequality at each finite M; no limiting approximation is substituted for a finite bound. In particular, for $M = p^a r^b$, it proves negativity above:

prime factors	sufficient modulus
$2, 3$	6^9
$2, r,\ r \geq 5$	$9\,000\,000$
$3 \leq p < r$	2^{22}

(5.1)

The sharper uniform cutoffs, apart from the four radicals $6, 10, 14, 15$, are

$$M \geq C(p), \quad C(2) = 384^2, \quad C(3) = 512^2, \quad C(p) = 600^2 \ (p \geq 5). \tag{5.2}$$

Appendix E specifies all finite domains and reductions. The certificates are organized by domains of moduli, not assumed experimental monotonicity of means.

Proof of Theorem 1.2(3). First the analytic bounds and the finite bridges settle the four exceptional radicals $6, 10, 14, 15$ at every pair of positive valuations. The squarefree two-prime moduli are settled by a single-mark reduction, with 4596 smaller moduli certified explicitly. For all bases, the analytic cutoffs leave three sixth powers, three fourth powers, 35 cubes and 99 squares, as detailed in the appendix. These imply the assertion for every length $N \geq 2$. The stronger length cutoffs are reindexings of the same modulus, not new hypotheses on states.

For $N = 1$, write $M = p^a r^b$. If $\gcd(a, b) \geq 2$, it is already a higher power of a two-prime base. Remove squarefree moduli, the four settled radicals, and the moduli above (5.2). The resulting exact inventory has 19239 profiles. A larger-prime-exponent-one lemma pays 7857 profiles. The five explicitly proved analytic bounds pay another 6972. The remaining 4410 profiles split into the three disjoint certificate domains of sizes 1087,1422 and1901. Each retained integer Q in (3.7) is strictly negative. All inventories, prime tests, powers, exclusions and sign comparisons are finite integer operations. The domain identities and mass totals are checked in the companion. There is no unexamined valuation class. □

6 A lower estimate and positive corrections

For squarefree q, put

$$D_q = 2^{\omega(q)}, \quad \rho_q = \prod_{p \mid q} \frac{p}{p+1}, \quad \alpha_q = \frac{2\rho_q}{\zeta(2)}, \quad \beta_q = -\frac{1 + \log \alpha_q}{2}.$$

Lemma 6.1. *With $q = \operatorname{rad}(M)$ and $L = \sqrt{M/\alpha_q}$,*

$$\mu(M) \geq \beta_q - \left(\frac{2(D_q + 3)}{\sqrt{\alpha_q M}} + \frac{1}{M} \right) \log L. \tag{6.1}$$

Proof. Let $P_q(T)$ count positive primitive pairs $a, d \leq T$ with $\gcd(d, q) = 1$. Möbius inversion gives

$$P_q(T) = \sum_{\substack{e \leq T \\ \gcd(e,q)=1}} \mu_{\mathrm{Mob}}(e)\, \lfloor T/e \rfloor\, U_q(\lfloor T/e \rfloor).$$

Inclusion–exclusion gives $|U_q(x) - (\varphi(q)/q)x| \le D_q$. Expanding each product, summing $e^{-1} \le 1 + \log T$, and bounding the tail of $\sum e^{-2}$ by $1/T$, yields

$$\left| P_q(T) - \frac{\rho_q}{\zeta(2)} T^2 \right| \le (D_q + 3)T(1 + \log T). \tag{6.2}$$

The main coefficient follows from $(\varphi(q)/q) \sum_{\gcd(e,q)=1} \mu_{\mathrm{Mob}}(e) e^{-2} = \rho_q/\zeta(2)$. These bounds also cover $T = 1$. Every admissible fraction can be reduced to a primitive one, so $C(T) \le 1 + 2P_q(T)$ by the union bound; injectivity is not needed. Use the resulting upper bound for $C(\lfloor x \rfloor)$ on $1 \le x \le L$ in $\mathbb{E} \log R_M = \int_1^\infty (1 - C(\lfloor x \rfloor)/M)\, dx/x$, and discard the nonnegative remainder. Integrating gives β_q as the main centered term. The error integral is at most $2(D_q + 3)L \log L/M + \log L/M$. The omitted lower endpoint main term is nonnegative, giving (6.1). □

For fixed B, this bound and (2.1) yield $E_{B,N}/N \to 1/2$, consistently with the antecedent in [3].

6.1 The exact finite obstruction

The direct height histogram for $M = 900$ has total mass 900 and height sum 34007. If $P = \prod_{s=0}^{899} R_{900}(s)$, integer multiplication gives

$$2 \cdot 30^{900} \le P < 3 \cdot 30^{900}. \tag{6.3}$$

The second bound is recorded for reproducibility, not needed for the sign. Hence $\mathbb{E} \log_{30} R_{900} = \log_{30} P/900 > 1$. An independent algorithm enumerates admissible denominators and centered numerators directly. Replacing full-unit denominators by denominators coprime to only one prime changes many minima; that wrong model is not used. The decimal $1.0002564533304775\ldots$ is descriptive; the proof is the integer comparison (6.3). It refutes a possible strengthening, not a conjecture stated in Question 3.2.

6.2 An eventually positive family

For $q_0 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 = 510510$,

$$\rho_{q_0} = 12155/41472.$$

For $M \ge q_0^2$, the main lower term in (6.1) satisfies $\beta_{q_0} > 7/500$, and its error is $< 13/1000$. We give rational checks to make this explicit. The elementary enclosures $e < 87/32$ and $\zeta(2) > 41/25$ imply

$$e\alpha_{q_0} < 1 - \frac{256169}{9068544}, \qquad \beta_{q_0} > \frac{256169}{18137088} > \frac{7}{500}.$$

The first follows from a factorial-series upper bound for e, the second from a partial sum for $\zeta(2)$. Also $\zeta(2) < 5/3$ gives $\alpha_{q_0} > 2431/6912 > 7/20 > (59/100)^2$. At $M = q_0^2$, $\log L < 14$ and $q_0 > 510000$, so

$$\left(\frac{2(128+3)}{\sqrt{\alpha_{q_0} M}} + \frac{1}{M} \right) \log L < \left(\frac{262}{(59/100)510000} + \frac{1}{510000^2} \right) 14 < \frac{13}{1000}.$$

The functions $\log(\sqrt{M/\alpha})/\sqrt{M}$ and $\log(\sqrt{M/\alpha})/M$ decrease on this range. Thus $\mu(M) > 1/1000$, proving the family in Theorem 1.3. No minimal radical or minimal cutoff is claimed. For radicals containing q_0, an eventually positive conclusion also follows from (6.1) with that radical's own constants.

7 The global additive optimum

Proof of Theorem 1.4. Let $k = \omega(q)$. Since the i-th smallest distinct prime is at least $i+1$, $q \geq (k+1)!$. Lemma 2.1 therefore gives $J(q) = q^{o(1)}$ uniformly over squarefree radicals: for $k \to \infty$,

$$\frac{\log J(q)}{\log q} \leq \frac{\log 2 + k \log 4}{\log((k+1)!)} \longrightarrow 0;$$

for bounded k and $q \to \infty$, the numerator is bounded. Equation (2.1) gives

$$c_+(B) \leq \frac{\log(J(\operatorname{rad} B)^2 + 3) + 1}{2 \log B}.$$

If the radical tends to infinity this goes to zero, since $\log q \leq \log B$. If radicals stay bounded their numerator is bounded and $\log B \to \infty$. This proves $c_+(B) \to 0$.

The witness $c(30) \geq E_{30,2} - 1 > 0$ now excludes all sufficiently large bases from the global supremum. Hence only finitely many bases remain. Proposition 2.3 supplies the common cap $C^* \leq \max(0, w, 39/40) < 49/50$. Together with the positive witness, this gives $0 < C^* < 49/50$. The sharper upper endpoint and the candidate lower endpoint in Theorem 1.4 follow from Section 8. A finite maximum of the quantities $c(B)$ is still a maximum of suprema over infinitely many lengths; finite-length attainment is not part of the conclusion. □

This elementary argument requires no hypothesis about extremal Jacobsthal gaps or prime distributions. It is not a practical algorithm for locating the maximizing base.

Corollary 7.1. *If $c(B) = C^*$, then B has at least three distinct prime factors and is not a perfect power. Equivalently, the gcd of its positive prime valuations is one.*

Proof. The classes with one or two primes have $c(B) \leq 0 < C^*$. For $B = A^e$, $e \geq 2$, equality of modulus and witness spaces gives

$$E_{A^e,N} - N/2 = \frac{1}{e}(E_{A,eN} - eN/2), \qquad c(A^e) \leq c(A)/e \leq C^*/e < C^*.$$

The supremum here is over a subset of the lengths for A; no maximizing finite length is asserted. Non-perfect-power does not mean squarefree. □

Corollary 7.2 (A classical Jacobsthal sharpening)**.** *Using Kanold's bound $J(q) \leq 2^{\omega(q)}$, as reported by Costello–Watts [5, Introduction, pp. 1–2], one has*

$$0 < C^* \leq \frac{\log 67 + 1}{2 \log 30} < \frac{5037}{6400} < \frac{4}{5}.$$

In particular, $E_{B,N} < N/2 + 4/5$ for all $B \geq 2, N \geq 1$.

Proof. The maximum cyclic gap $J(q)$ equals the least length of a consecutive integer interval guaranteed to contain a q-unit: a gap of length G has $G-1$ intervening nonunits. Thus the cited bound uses precisely our convention, and applies to every radical, not only primorials.

Put $q = \operatorname{rad}(B)$ and $k = \omega(q)$. For $k \leq 2$, Theorem 1.2(2)–(3) gives $c(B) \leq 0$. For $k \geq 3$, equation (2.1), uniformly in N, gives

$$c(B) \leq \frac{\log(4^k + 3) + 1}{2 \log q}.$$

The first three distinct prime divisors have product at least 30, and each further prime is at least 7, hence at least 4. Consequently $q \geq 30\, 4^{k-3}$. Also $\log(4^k + 3) \leq k \log 4 + \log(67/64)$, so

$$c(B) \leq F(k) := \frac{k \log 4 + 1 + \log(67/64)}{2 \log 30 + 2(k-3) \log 4}.$$

The denominator is positive for real $k \geq 3$; the numerator of its derivative is $2\log 4(\log 30 - \log 67 - 1) < 0$. Thus $F(k) \leq F(3) = (\log 67 + 1)/(2\log 30)$. This is a common cap before either supremum over N or B.

For a rational enclosure, Appendix C gives $\log 2 < 7/10$ and $e < 11/4$. Together with $\log(1+x) < x$,

$$\log 67 + 1 = 6\log 2 + \log(67/64) + 1 < \frac{21}{5} + \frac{3}{64} + 1 = \frac{1679}{320}.$$

Since $11^{10} = 25937424601 < 28311552000 = 27000\,4^{10}$, one has $e^{10} < 30^3$, hence $\log 30 > 10/3$. Therefore

$$\frac{\log 67 + 1}{2\log 30} < \frac{1679/320}{20/3} = \frac{5037}{6400} < \frac{5120}{6400} = \frac{4}{5}.$$

Positivity follows from Theorem 1.3. □

This sharpening imports the cited classical gap bound; the self-contained $49/50$ argument above remains valid independently. The argument does not determine the optimal correction or its extremizers.

8 A sharper uniform additive correction

Write $\Delta(B, N) = \mu(B^N)/\log B$, in the model of Section 1. The following strengthens the bound on the same constant C^*. The elementary $49/50$ bound and the classical $4/5$ consequence remain separate proofs and do not require the certificates used here.

Theorem 8.1. *For every pair of integers $B \geq 2$ and $N \geq 1$,*

$$\Delta(B, N) \leq U_* < 0.241933621228105, \qquad U_* = N_*/D_*.$$

The exact numerator and denominator are

$$\begin{aligned} N_* = {} & 40623994785018517492427 \cdot 10^{75} \\ & + 23261169875014398280856 52 \cdot 10^{50} \\ & + 36010108438519239477952 80 \cdot 10^{25} \\ & + 68898373462485692965394 51, \\ D_* = {} & 16791380453366837429769 5 \cdot 10^{75} \\ & + 20152035872652169461925 75 \cdot 10^{50} \\ & + 10026452960100000000000 00 \cdot 10^{25}. \end{aligned}$$

Consequently $0 < C^* < 0.241933621228105$.

The upper bound combines inequalities valid for infinitely many moduli with finite integer and rational certificates. We give the inequalities, their quantifiers and the finite predicates below. A certificate of an admissible upper witness need not certify the minimum. A recorded successful execution, or agreement of two reported constants, does not replace the finite predicates.

8.1 Shortest vectors and a coupled shear bound

Let λ_s be the Euclidean minimum of Λ_s. Two distribution bounds needed here are, for every integer $M \geq 1$ and real $x \geq 0$,

$$\mathbb{P}(\lambda_s^2/M \leq x) \leq \min\left(1, \frac{3}{2}x,\ ax + b\sqrt{x/M} + \frac{d}{M}\right), \tag{S1}$$

where

$$a = \frac{6414336}{5010005}, \qquad b = \frac{45173053}{51531480}, \qquad d = \frac{221184}{715715}.$$

The $M = 1$ case is included because a transferred profile can have that seed. These are bounds on an unmarked Euclidean minimum, not the distribution of the marked height R_M. Appendix F proves them.

Let H_k be a length for which every interval of H_k consecutive integers contains a point avoiding one arbitrary residue class modulo each of at most k distinct primes. The following values suffice:

k	1	2	3	4	5	6	7	8	9	10	11	12	13	14
H_k	2	4	6	10	14	22	26	44	58	66	98	148	204	360.

(S2)

They are upper bounds for all such supports; no assertion that primorials give the worst support is used. The finite covering predicates and their infinite-support implication are proved in Appendix G.

Lemma 8.2. *Suppose the above avoidance property holds with an even length $H = 2h$ for at most $\omega(M) - 1$ primes. Then, for every state,*

$$R_M(s) \le \kappa(H)\frac{M}{\lambda_s}, \qquad \kappa(H)^2 = 1 + \max\left(\frac{H^2}{4}, \frac{(H-1)^2}{3}\right). \tag{S3}$$

Proof. Complete a shortest lattice vector u to a basis u, v, reducing $v = \theta u + v_\perp$ with $|\theta| \le 1/2$. Put $\lambda = \|u\|_2$. Minimality gives

$$\lambda^4(1-\theta^2) \le M^2, \qquad \|v_\perp\|_2 = M/\lambda.$$

The vertical projection of the lattice is $\mathbb{Z}$, hence $\gcd(u_d, v_d) = 1$. If u_d is a full unit, the vector u suffices, since $\lambda^2/M \le 2/\sqrt{3} < \kappa(H)$. Otherwise some prime divisor of M divides u_d; it does not divide $v_d + ju_d$ for any j. Each remaining prime forbids one class of j. The allowed integers thus have consecutive gaps at most $2h$.

Put $t = |\theta|$. The integer structure gives an allowed j with $|j + \theta| \le h - t$. Indeed, after reflection assume $\theta = t$. If zero is allowed use it. Otherwise let $-l, r$ be the allowed neighbors of zero, where $l, r \ge 1$, $l + r \le 2h$. If $l \le h$, use $-l$; if $l \ge h + 1$, use $r \le h - 1$. Therefore

$$\frac{\|v + ju\|_2^2}{(M/\lambda)^2} \le 1 + \frac{(h-t)^2}{1-t^2} \le 1 + \max\left(h^2, \frac{(2h-1)^2}{3}\right).$$

For $h = 1, 2, 3$ the last inequality follows from $h^2(1-t^2) - (h-t)^2 = t[2h - (h^2+1)t] \ge 0$. For $h \ge 4$ it follows from

$$(2h-1)^2(1-t^2) - 3(h-t)^2 = (1-2t)[2(h^2 - h + 1)t + h^2 - 4h + 1] \ge 0.$$

The full-unit denominator is nonzero; choose its sign positive. Finally $\|\cdot\|_\infty \le \|\cdot\|_2$. □

8.2 Positive and negative logarithmic contributions

Set

$$Z_M = R_M/\sqrt{M}, \quad P(M) = \mathbb{E}(\log Z_M)_+, \quad L(M) = \mathbb{E}(-\log Z_M)_+, \quad \mu(M) = P(M) - L(M).$$

For κ as in (S3), any rational $z > 1$ with $z < \kappa\sqrt{M}$ gives

$$P(M) \le C(M, \kappa, z) := \log z + \frac{a\kappa^2}{2z^2} + \frac{b\kappa}{z\sqrt{M}} + \frac{d}{2M}\log\frac{\kappa^2 M}{z^2} - \frac{a/2 + b}{M}. \tag{S4}$$

To see this, use (S1) and (S3) above z, with the trivial probability bound below z. The witness tail is empty beyond $\kappa\sqrt{M}$, since $\lambda_s \geq 1$. Integrating the three terms to that finite endpoint gives (S4), including both endpoint subtractions. In particular the constant d/M is not integrated to infinity.

For a scalar threshold Q, let $E_0 = a/2 + b$. The finite conditions

$$\log(\kappa^2 Q/z^2) > 1, \qquad b^2\kappa^2 Q > 4z^2 E_0^2 \tag{S5}$$

prove $C(M, \kappa, z) \leq C(Q, \kappa, z)$ for every real $M \geq Q$. Indeed

$$M^2 \partial_M C = E_0 - \frac{b\kappa\sqrt{M}}{2z} + \frac{d}{2}\left(1 - \log\frac{\kappa^2 M}{z^2}\right) < 0.$$

The cuts in the certificate are rationals on the grid 10^{-6}; their choice can be reproduced by rounding upward the positive root of

$$(1 - d/Q)z^2 - b\kappa z/\sqrt{Q} - a\kappa^2 = 0.$$

Only (S5) at the actual rational cut is a proof obligation; exact stationarity is unnecessary.

For the negative part, fix once and for all

$$\mathcal{P} = \{2, 3, 5, 7, 11, 13, 17, 19\}, \quad S = \mathcal{P} \cap \operatorname{supp}(M), \quad P_S = \prod_{p \in S} p.$$

Appendix H constructs two nondecreasing rational profiles $h^{1/2}_{w,S}$, $h^{4/5}_{w,S}$, for $w = \omega(M)$. The first counts primitive full-unit vectors in a square. The second, used for $w = 5, 6$, counts them in the cross

$$\{|a| \leq R,\ 0 < d \leq \lfloor R/2 \rfloor\} \ \cup\ \{|a| \leq \lfloor R/2 \rfloor,\ 0 < d \leq R\}.$$

Their injectivity conditions are respectively $2R^2 < M$ and $5R^2 < 4M$, both strict. Define

$$r_\alpha(Q) = \max\{r \in \mathbb{Z}_{\geq 0} : r^2 < \alpha Q\}, \qquad G^\alpha_{w,S}(u; Q) = \begin{cases} u\, h^\alpha_{w,S}(\lfloor\sqrt{uQ}\rfloor), & 0 < u < \alpha, \\ \alpha h^\alpha_{w,S}(r_\alpha(Q)), & \alpha \leq u \leq 1. \end{cases} \tag{S6}$$

For every modulus $M \geq Q$ in that class, $\mathbb{P}(Z_M \leq \sqrt{u}) \geq G^\alpha_{w,S}(u; Q)$. Consequently put

$$I_{4,S}(Q) = \frac{1}{2}\int_0^1 G^{1/2}_{4,S}(u; Q)\frac{du}{u}, \qquad I_{w,S}(Q) = \frac{1}{2}\int_0^1 \max(G^{1/2}_{w,S}, G^{4/5}_{w,S})(u; Q)\frac{du}{u} \quad (w = 5, 6). \tag{S7}$$

Then $L(M) \geq I_{w,S}(Q)$. The two regions overlap: their counts are not added. The maximum is taken between lower bounds on the same physical CDF. The support S is fixed through the entire integral.

8.3 Exhaustive partition of supports

The auxiliary certified estimates in Appendix I give $\Delta(B, N) \leq 2/85$ for $\omega(B) = 3$, and for the following four-prime families other than the first, with every positive valuation and every $N \geq 1$:

q	30	42	70	66	105	78	110	102	154	165	130	114
$p_{\min}$	7	11	11	13	11	17	13	19	13	13	17	23.

(S8)

A column denotes $\operatorname{rad}(B) = qp$, $p \geq p_{\min}$ prime. The $30p$ column has the bound

$$\Delta(B, N) \leq c_0 := \Delta(210, 1), \qquad 0.023672606561106 < c_0 < 0.023672606561107. \tag{S9}$$

No characterization of equality is needed. The one- and two-prime results already proved give $\Delta \le 0$. The five-prime supports separately bounded by $2/85$ are

$$\mathcal{E} = \{2310, 2730, 3570, 3990, 4290, 4830, 5610, 6006, 6090, 6270, 6510\}. \tag{S10}$$

The negative profiles for $w = 5$ are conditional on $\mathrm{rad}(M) \notin \mathcal{E}$; they do not prove the exclusions themselves.

The smallest remaining four- and five-prime radicals are respectively $Q_4 = 3003$ and $Q_5 = 6630$. There are exactly 84 squarefree four-prime integers at most 3003, of which 83 lie in (S8), and exactly 12 squarefree five-prime integers at most 6630, of which eleven lie in $\mathcal{E}$. These finite inventories can be checked either by factorization or by ordered prime tuples. The prime limits 100 and 31 suffice: the product of the other $w-1$ distinct primes is at least their primorial. This elementary use of primorials is a size bound, not an extremal assumption for rational complexity. For six primes use $Q_6 = 30030$.

In a residual class with $w = 4, 5, 6$, put $t = w - |S|$. The ordered external primes exceed 19. Lower-bound them coordinatewise by $\ell_{S,j}$, the first t primes above 19, except that when $w = 5, t = 1$ the first lower bound is the least prime $r > 19$ with $P_S r \notin \mathcal{E}$. Omit a class with $t = 0$ precisely when $w = 5$ and $P_S \in \mathcal{E}$. Define

$$Q_{w,S} = \max\left(Q_w, P_S \prod_{j=1}^{t} \ell_{S,j}\right). \tag{S11}$$

Every actual base in the residual class satisfies $B \ge Q_{w,S}$, hence $M = B^N \ge Q_{w,S}$. For example $w = 5, S = \{2, 3, 5, 7\}$ has $\ell_{S,1} = 37$ and $Q_{5,S} = 7770$. The skipped primes $23, 29, 31$ give exactly three excluded radicals. For $t \ge 2$, ignoring extra exclusions only weakens the lower bound. There are respectively 163, 211 and 247 admitted classes.

For each class use $H = H_{w-1}$, the corresponding κ, the cut $z_{w,S}$ and its own $Q_{w,S}$. Equations (S4)–(S11) yield

$$\Delta(B, N) \le \frac{\max\{0, C(Q_{w,S}, \kappa, z_{w,S}) - I_{w,S}(Q_{w,S})\}}{\log Q_{w,S}}. \tag{S12}$$

The sign check precedes replacing $\log B$ by its lower bound. All three ingredients retain the same S; only after computing complete normalized rows is their maximum taken. The 621 rational interval comparisons give the outward sector bounds

w	upper bound
4	0.202966987434704
5	0.241933621228105
6	0.241858537967210.

Each displayed decimal rounds up a rational upper bound; the exact largest rational is U_* in Theorem 8.1. The rows producing the largest upper bounds are analytical bounds, not identified optimizing physical bases.

8.4 Seven or more prime factors

For $7 \le w \le 15$, use Q equal to the w-th primorial, $\kappa(H_{w-1})$, a cut satisfying (S5), and $L(M) \ge 0$. The certified outward bounds are

w	$\Delta \le$	w	$\Delta \le$	w	$\Delta \le$
7	.237632277858435	10	.182279486252701	13	.151871560366695
8	.204863443985275	11	.163269471241710	14	.145175831262366
9	.199567498654016	12	.156877570209516	15	.145447882175961

(S13)

The companion stores the exact fractions and cuts.

For the unbounded range, use the classical inequality $J(q) \le 2^{\omega(q)}$ of Kanold, in the formulation reported by Costello–Watts [5, Introduction]. It is an imported classical theorem, not a consequence of the finite covering table. Let P_{16} be the sixteenth primorial, whose largest prime is 53, and write $v = w - 16 \ge 0$. Set $H_0 = 2^{15}$, $\kappa_0^2 = (H_0^2 + 3)/3$, $\kappa_v = 2^v \kappa_0$, $z_v = 2^v z_0$, $Q_v = P_{16} 59^v$. The classical geometry gives a witness factor at most κ_v, because

$$(H_0^2\, 4^v + 3)/3 \le 4^v (H_0^2 + 3)/3.$$

Here the weaker classical factor is retained deliberately: the improvement $H - 1$ in (S3) has not been assumed to obey this scaling inequality. Since $B \ge Q_v$,

$$\Delta(B, N) \le \frac{C(P_{16}, \kappa_0, z_0) + v \log 2}{\log P_{16} + v \log 59}. \tag{S14}$$

This uses the identity $C(Q_v, \kappa_v, z_v) = v \log 2 + C(Q_v, \kappa_0, z_0)$ and the real-M derivative check at P_{16}. The derivative of the last ratio has the sign of

$$(\log 2)(\log P_{16}) - C(P_{16}, \kappa_0, z_0) \log 59,$$

whose certified upper endpoint is negative. Thus its value at $v = 0$ bounds all real $v \ge 0$, and in particular all unbounded integer cardinalities. It is less than $0.233058795421166 < U_*$. This completes the proof of Theorem 8.1: the classes of (S12), (S13), (S14) and all excluded sectors form an exhaustive partition, and each upper bound is at most U_*. The positivity of C^* follows from the earlier counterexample, or from (S9).

9 Reproducibility and limitations

The repository companion contains the original finite mathematical premises, portable reconstruction programs, exact logarithmic sums, complete domain lists and a provenance map. The original default checks remain a short retained-evidence check, a fixed defining oracle and a four-case squarefree reconstruction. They are distinct from the complete reconstruction route provided by `scripts/reconstruct_all.py`. The reproduction guide specifies the commands, domains and limits of both routes.

The complete source-replay chain passed all 37 stages in each of normal and optimized Python modes. The squarefree prefix, which originally retained only digests and signs, now has all 4596 reconstructed upper histograms and their consumed logarithmic ledgers. These histograms cover 41585055 original envelope entries. A separate K32 logarithm implementation checked all 705257 consumed tick entries, involving 6929 distinct integers, and reconstructed every negative outward sum and its exact normalization. The retained full artifacts are indexed by `companion/reconstruction/index.json`; `scripts/check_complete_evidence.py` checks their declared complete coverage, byte identities and the squarefree logarithmic certificates.

An additional distinct central checker reconstructs all integer thresholds $1 \le T \le k = \lfloor \sqrt{M} \rfloor$ and every proper original gcd stratum for all nine aggregate batches, including their consumed validation rows. For each full-unit denominator $1 \le d \le k$, it enumerates every $1 \le a \le k$ and both signed residues $\pm a d^{-1} \pmod M$, retaining the smallest $\max(a, d)$ at each state. There is no $\gcd(a, d) = 1$ filter: nonprimitive pairs are legal witnesses too. Zero is one separate atom; coincident signed witnesses update the same state. The resulting central height bins give every stratum prefix. All source-produced central arrays are compared at every threshold, including explicit zero counts for omitted strata above k. The original endpoint-only checker descriptions elsewhere in the paper refer to the earlier retained-data checks, rather than this additional complete central comparison.

The same direct central enumeration also checks every central histogram prefix for all 4596 squarefree cases, the eight base-six cases and the three sixth-power cases. All nine aggregate and all three pointwise-domain checks completed in both normal and optimized modes. The pointwise histograms provide global prefix comparison targets; they do not retain separate source stratum arrays. These checks compute exact central minima, not complete true minimum vectors in the far range. Far-envelope domination, the infinite estimates and the reduction arguments still use the proved mathematics and the source producer. The attributed binary mean theorem remains an import; this reconstruction does not repeat its earlier paper.

The sharper correction adds the companion in `companion/global_bound`. Its scalar producer and distinct checker recompute all 621 supportwise rows and the remaining cardinality tail. Its premise checker recomputes the incidence period, residue-class covering predicates, shear comparisons and exact candidate minimum histogram. Retained closure checks validate the compact numerical premises and their exact domain joins. Full C++ producer/consumer routes reconstruct the geometric and finite witnesses. The early closure reconstructions were observed on 8 October 2026, with their numerical kernels and inputs preserved; later closure components were reconstructed for this revision. These scopes are separate from the unchanged 37-stage two-prime chain. A retained-only check does not mean that a geometric producer was rerun. Full routes require GNU C++17, `__int128`, OpenMP and little-endian Linux. Missing carrier members have been reconstructed with new provenance; historical hash agreement is recorded only where actually verified.

The paper proves mean bounds for its full-unit rational-height model. It does not determine the optimal correction, maximizing lengths or a sign classification for all bases with at least three primes. It makes no assertion about the Collatz conjecture. The classical tools and earlier binary theorem are attributed, and no universal priority claim is made.

Data and code availability. All author-generated integer certificates and programs needed for the finite proof dependencies accompany the paper in the repository. The manuscript is licensed under CC BY 4.0; original code, certificates and repository documentation are licensed under MIT. Cited publications retain their own rights and are not bundled.

AI-use disclosure. OpenAI GPT models assisted with mathematical exploration, proof criticism, literature searches, manuscript editing, implementation and reproducibility checks. Several AI review passes shared source exposure. These do not constitute independent human peer review, formal proof verification or an originality certificate. The author retains responsibility for the arguments, source attributions and accompanying artifacts.

A A finite compensation inequality for two-prime moduli

Let $M = p^a r^b$, where $p < r$ are distinct primes and $a, b \geq 1$. Throughout this appendix the average is over all M residues, including zero, and

$$R_M(s) = \min_{\substack{d>0,\ (d,M)=1 \\ A \equiv sd \pmod M}} \max(|A|, d), \qquad \mu(M) = \frac{1}{M} \sum_{s \bmod M} \log R_M(s) - \frac{1}{2} \log M.$$

All logarithms are natural. Write

$$A_0 = \sqrt{M}, \quad D = pr, \quad u = (1 - p^{-1})(1 - r^{-1}),$$
$$\rho = \frac{pr}{(p+1)(r+1)}, \quad \alpha = \zeta(2)^{-1}, \quad \ell = \log 2.$$
$$\kappa = \alpha\rho, \qquad \theta = \alpha(1-\rho), \qquad \beta = 3 - 2\sqrt{2}.$$

The symbol A_0 denotes a real scale, not a numerator.

A.1 Unit discrepancy and primitive counting

Define

$$U(x) = \lfloor x \rfloor - \lfloor x/p \rfloor - \lfloor x/r \rfloor + \lfloor x/D \rfloor, \qquad e(n) = U(n) - un.$$

Let $h = \max_{0 \le n < D} e(n)$, $m = \min_{0 \le n < D} e(n)$, and put

$$K = h - m + u, \qquad W = \max(h, -m + 2u), \qquad H_0 = u + h.$$

Then $h \ge 0$, $m \le 0$, and $H_0 \ge 1$, the last inequality following from $e(1) = 1 - u$. The real discrepancy lies in $[m - u, h]$. Every closed interval of length v in the positive half-line therefore contains at most $uv + K$ positive D-units: for its integer endpoints $i = \lceil x \rceil$, $j = \lfloor y \rfloor$, use $U(j) - U(i-1)$ and $j - i \le y - x$. This includes both endpoint atoms and zero-width intervals. Partial summation gives

$$\sum_{\substack{d \le Q \\ (d,D)=1}} \frac{1}{d} \le H_0 + u \log Q \quad (Q \ge 1).$$

For $x = n + t$, $0 \le t < 1$,

$$nU(n) - ux^2 = ne(n) - u(2nt + t^2), \qquad |nU(n) - ux^2| \le Wx.$$

For the lower bound one uses

$$(-m + 2u)x - \{-nm + u(2nt + t^2)\} = 2un(1 - t) + (-m + 2u)t - ut^2 \ge 0;$$

the upper bound follows from $h \ge 0$.

Let $P(T)$ count pairs $1 \le A, d \le T$ with $(A, d) = 1$ and $(d, D) = 1$. For integer $T \ge 1$, cancelling content only at D-units yields

$$P(T) = \sum_{\substack{g \le T \\ (g,D)=1}} \mu_{\mathrm{Mob}}(g) \lfloor T/g \rfloor U(\lfloor T/g \rfloor).$$

The Euler product

$$u \sum_{(g,D)=1} \frac{\mu_{\mathrm{Mob}}(g)}{g^2} = \frac{u\alpha}{(1 - p^{-2})(1 - r^{-2})} = \kappa$$

and the integer-cutoff bound $\sum_{g > T} g^{-2} \le T^{-1}$ prove

$$|P(T) - \kappa T^2| \le T(b_0 + b_1 \log T), \tag{1}$$

with either of the *coherent* choices

$$(b_0, b_1) = (W + u, W) \quad \text{or} \quad (b_0, b_1) = (WH_0 + u, Wu).$$

Both have $b_0 \ge b_1$. The elementary bounds $h < 2$, $m > -1$ also permit the coarse pair $(b_0, b_1) = (4, 3)$ and $K = 4$: here $|nU(n) - ux^2| \le 3x$, and the unmarked harmonic sum is at most $1 + \log T$.

For later sharpening we need the stronger uniform integer discrepancy

$$-u \le e(n) \le 2u. \tag{2}$$

It suffices to work in one period. At units $e(n) - e(n-1) = 1 - u > 0$, so a minimum occurs at zero or a nonunit. If $n = pt < D$, write $pt = jr + s$, with $1 \le t < r$, $0 \le j \le p - 1$, $1 \le s < r$. Then $e(pt) = (s - t)/r$, and

$$t - s = jr - (p - 1)t \le \frac{jr - (p - 1)}{p} \le \frac{(p - 1)(r - 1)}{p} = ur.$$

Interchanging the primes treats multiples of r. Reflection gives $e(D-n) = \mathbf{1}_{(n,D)=1} - e(n)$ for $0 < n < D$. At nonunits this implies $e(n) \le u$. At a unit t, the lower bound for $e(t-1)$ implies $e(t) \ge 1 - 2u$; reflection then gives $e(n) \le 2u$. Thus $K \le 4u$, $W \le 3u$. For the second coherent parameter choice,

$$b_0 \le 9u^2 + u, \quad b_1 \le 3u^2, \quad b_0 - b_1 = Wh + u \le 6u^2 + u. \tag{3}$$

A.2 Central collisions with closed endpoints

For integer $T^2 \le M$, let $C(T) = \#\{s : R_M(s) \le T\}$. Every nonzero short witness can be cancelled to a primitive one because its content is a unit modulo M. Two primitive witnesses with the same numerator sign represent different states unless they are identical: a nonzero determinant would be a multiple of M, while its magnitude is $< M$. At $T^2 = M$, equality in this bound would require a denominator T, which is not a unit. Opposite signs can meet only when

$$Ad + Eb = M, \qquad 1 \le A, E, b, d \le T, \quad (bd, D) = 1.$$

The value $2M$ is impossible for the same nonunit-endpoint reason. There is at most one witness of either sign at a given state. Neither a nonzero short witness nor its negative can be the zero state. Consequently

$$C(T) = 1 + 2P(T) - J(T), \tag{4}$$

where $J(T)$ counts the ordered positive/negative collision pairs above; no further factor two is inserted.

A common divisor of b, d divides M and is coprime to M, so $(b, d) = 1$. For fixed b, d, the interval of progression parameters has width at most

$$\frac{T(b+d) - M}{bd} < 1$$

when nonempty. Indeed $T(b+d) - M - bd \le -(T-b)(T-d)$, with equality requiring $M = T^2$ and one nonunit denominator T. Hence at most one collision occurs per ordered denominator pair. Put $L_T = (2T - M/T)_+$. The deficits $T - b, T - d$ lie in the closed triangle $x + y \le L_T$. If $V(v)$ counts admissible deficits up to v, then $V(v) \le uv + K$. Summing the linear weights as a Stieltjes integral gives

$$J(T) \le \frac{u^2 L_T^2}{2} + 2uKL_T + K^2 \quad (2T^2 > M). \tag{5}$$

There are no collisions for $2T^2 < M$. If $2T^2 = M$, equality forces $b = d = T$; the odd prime divisor of M divides T, so these are not units. Thus $J(T) = 0$ also at this boundary.

A.3 The global far cover

For integer $T^2 \ge 2M$, consider the complete Farey sequence of order T. A unit-denominator center c/d covers the closed interval $[c/d - T/(Md), c/d + T/(Md)]$. Consecutive centers $a/q < c/d$ have $cq - ad = 1$, $(q, d) = 1$, and $q + d > T$. If both denominators are units their intervals overlap. If only one is a unit, an uncovered point forces the other denominator to be $< M/T$.

If both are nonunits, their two prime obstructions are complementary. Hence $q - d$ is a unit modulo D. On the Farey arc the affine residual

$$(q-d)x - (a-c)$$

has values $1/q$ and $1/d$ at its endpoints. If $q, d \geq M/T$, its absolute value multiplied by M is at most T, and the vector difference is a genuine witness of denominator $0 < |q-d| \leq T$ on the whole arc. A witness center outside that arc is allowed; no isolation of deleted centers is presumed.

It follows that every uncovered point belongs to an arc incident to a deleted reduced center a/q, $0 < a < q$, with $q < M/T \leq T/2$. A complete Farey neighbor c/d has $d > T - q > q$. If d is also a nonunit, replace it by $(c-a)/(d-q)$. Its denominator is a unit, lies in $(T-2q, T]$, and has determinant one with a/q on the same side. If $D \mid q$, the original neighbors are already units and lie in $(T-q, T]$. The uncovered residues are therefore contained in the union of open holes

$$\left(\frac{Ma}{q} - \frac{M-qT}{qb_L}, \frac{Ma}{q} + \frac{M-qT}{qb_R}\right), \qquad b_L, b_R > T - 2q.$$

An open interval of length v contains at most $v+1$ integers. Overlapping holes may be counted repeatedly in an upper bound. Consequently

$$M - C(T) \leq \sum_{\substack{q<M/T \\ (q,D)>1}} \varphi(q) \left\{ \frac{2(M-qT)}{q(T-2q)} + 1 \right\}. \tag{6}$$

The $+1$ allowances, positive denominator marks and open endpoints are essential. The displayed cover holds also at $T^2 = 2M$, but a later partial summation must start strictly above that threshold.

A.4 Deleted-center density and its real-cutoff error

Put $A(Q) = \sum_{q \leq Q,\, (q,D)>1} \varphi(q)/q$, and define

$$v = 2 - p^{-1} - r^{-1}, \qquad c_0 = \frac{p-1}{p^2} + \frac{r-1}{r^2} - \frac{D-1}{D^2}.$$

Expanding $\varphi(q)/q = \sum_{d \mid q} \mu_{\mathrm{Mob}}(d)/d$ and separating its p, r parts yields the exact identity

$$A(Q) = \sum_{\substack{d \leq Q \\ (d,D)=1}} \frac{\mu_{\mathrm{Mob}}(d)}{d} \left\{ (1-p^{-1}) \left\lfloor \frac{Q}{pd} \right\rfloor + (1-r^{-1}) \left\lfloor \frac{Q}{rd} \right\rfloor - (1-D^{-1}) \left\lfloor \frac{Q}{Dd} \right\rfloor \right\}.$$

The linear coefficient in braces is $c_0 Q/d$, and their error is at most v in absolute value. Since $c_0 = (1-p^{-2})(1-r^{-2}) - u$, the leading density is θ. For real $Q \geq 1$, not merely integer Q,

$$Q \sum_{d > Q} d^{-2} \leq Q/\lfloor Q \rfloor < 2.$$

Thus

$$|A(Q) - \theta Q| \leq E(Q) = E_0 + E_1 \log Q \tag{7}$$

with either $E_0 = v + 2c_0, E_1 = v$, or $E_0 = vH_0 + 2c_0, E_1 = vu$. The coarse admissible choice is $E(Q) = 3 + 2\log Q$, since $c_0 \leq 17/36$. For partial summation on $0 \leq Q < 1$, take $E(Q) = E_0$: then $A(Q) = 0$, $\theta Q \leq \theta < 1/3$, and $E_0 \geq vu \geq 7/18$. A second partial summation gives

$$B(Q) := \sum_{\substack{q \leq Q \\ (q,D)>1}} \varphi(q) \leq \frac{\theta Q^2}{2} + 2QE(Q). \tag{8}$$

For $T^2 > 2M$, write $Q = M/T$ and $f(x) = (M - Tx)/(T - 2x)$. It decreases from M/T to zero at Q. Equations (6)–(8) imply

$$\frac{M - C(T)}{M} \le V(T/A_0) + \frac{4E(Q)}{T}, \qquad V(t) = \theta F(t/\sqrt{2}) + \frac{\theta}{2t^2},$$

where

$$F(z) = \sum_{n\ge 1} \frac{z^{-2n}}{n(n+1)}, \qquad F(1) = 1.$$

Indeed $2\theta M^{-1} \int_0^Q f(x)\,dx = \theta F(T/(\sqrt{2}A_0))$; the weighted discrepancy costs $2E(Q)/T$, and the integer-hole allowance costs another $2E(Q)/T$. The function V is decreasing, $V(\sqrt{2}) = 5\theta/4$, and termwise integration of its positive series gives

$$\int_{\sqrt{2}}^{\infty} V(t)\frac{dt}{t} = \theta\left\{\frac{\zeta(2) - 1}{2} + \frac{1}{8}\right\} = \frac{1-\rho}{2} - \frac{3\alpha(1-\rho)}{8}. \tag{9}$$

A.5 The complete finite compensation

Denominator one bounds every height by $\max(1, \lfloor M/2 \rfloor)$. The layer-cake identity is

$$\mu(M) = -D_c + P_{\text{mid}} + P_{\text{far}},$$

with

$$D_c = \frac{1}{M}\int_1^{A_0} C(\lfloor x \rfloor)\frac{dx}{x}, \qquad P_{\text{mid}} = \frac{1}{M}\int_{A_0}^{\sqrt{2}A_0} [M - C(\lfloor x \rfloor)]\frac{dx}{x}.$$

The floor lemma

$$\int_1^{A_0} \lfloor x \rfloor^2 \frac{dx}{x} \ge A_0^2/2 - A_0$$

follows by summing $\log(1 + 1/n) \ge 1/n - 1/(2n^2)$ and using $\log((k+t)/k) \ge t/(k+t)$ on the last interval. The residual comparison is at least $1/2 + t - 3t^2/2 \ge 0$ for $0 \le t < 1$. Integration of (1) and (5), retaining the closed collision boundary, gives

$$D_c \ge \kappa - u^2(3/4 - \ell) - \frac{2\kappa + 2(b_0 - b_1) + 2b_1 \log A_0 + 2uK\beta}{A_0} - \frac{K^2\ell}{2A_0^2}.$$

Here the two collision integrals are

$$\frac{1}{M}\int_{A_0/\sqrt{2}}^{A_0} (2x - M/x)^2 \frac{dx}{x} = 3/2 - 2\ell, \qquad \int_{A_0/\sqrt{2}}^{A_0} (2x - M/x)\frac{dx}{x} = A_0\beta.$$

The favorable central zero and favorable primitive-error endpoint $2(b_0 - b_1)/A_0^2$ have only been discarded. At $k = \lfloor A_0 \rfloor$, $k^2 \ge A_0^2 - 2A_0$. CDF monotonicity throughout the middle interval implies

$$P_{\text{mid}} \le (1 - 2\kappa + u^2/2)\frac{\ell}{2} + \frac{\ell(4\kappa + 2b_0 + 2b_1 \log A_0 + 2uK)}{2A_0} + \frac{K^2\ell}{2A_0^2}.$$

Start the discrete far sum at

$$L = \lfloor \sqrt{2M} \rfloor + 1.$$

The initial interval $[\sqrt{2M}, L]$ costs at most $1/(\sqrt{2}A_0)$, including when $\sqrt{2M}$ is an integer. This prevents the false substitution $f(Q) = 0$ at $T^2 = 2M$. For all subsequent terms $T^2 > 2M$. Use $\log(1 + 1/T) \le 1/T$, decreasing-sum comparison and $\sum_{T\ge L} T^{-2} \le L^{-1} + L^{-2}$. Equations (9) and (7) then give

$$P_{\text{far}} \le \frac{1-\rho}{2} - \frac{3\alpha(1-\rho)}{8} + \frac{1 + 5\theta/4 + 4E(A_0)}{\sqrt{2}A_0} + \frac{2E(A_0)}{A_0^2}.$$

When $Q < 1$ the survival is zero; no negative logarithmic error is used. Combining the three bounds proves

$$\mu(M) \le -\Delta(p,r) + \mathcal{E}_{p,r}(A_0), \tag{10}$$

where

$$\Delta = \rho\{\alpha(5/8+\ell)+1/2\} + 3\alpha/8 - 1/2 - \ell/2 - \tfrac{3}{4}(1-\ell)u^2$$

and

$$\begin{aligned}\mathcal{E}_{p,r}(A_0) = \frac{1}{A_0}\big[&2\kappa + 2(b_0-b_1) + 2b_1\log A_0 + 2uK\beta \\ &+ \tfrac{\ell}{2}(4\kappa + 2b_0 + 2b_1\log A_0 + 2uK) + (1+5\theta/4+4E(A_0))/\sqrt{2}\big] \\ &+ \frac{K^2\ell + 2E(A_0)}{A_0^2}.\end{aligned} \tag{11}$$

All these estimates hold for arbitrary positive prime valuations. They are inequalities for the original finite average, not limiting-law or monotonicity assertions.

A.6 Uniform leading margins and universal cutoffs

The margin Δ increases with either prime. To check this algebraically put $x = 1/p, y = 1/r$, $A_* = \alpha(5/8+\ell)+1/2$, $C_* = \frac{3}{4}(1-\ell)$. Multiplication of $\partial_x\Delta$ by $(1+x)/\rho$ gives

$$-A_* + 2C_*(1-x)(1+x)^2(1-y)^2(1+y) < 0.$$

For $0 \le x, y \le 1/2$, its positive term is $< 9/8$, whereas $A_* > 51/40$, using $\alpha > 3/5$ and $\ell > 2/3$. Thus both reciprocal-prime derivatives are negative.

The coarse coherent parameters $(b_0, b_1, K) = (4, 3, 4)$ and $E(A_0) = 3 + 2\log A_0$ give

$$\mathcal{E}_{p,r}(A_0) < \frac{22 + 15\log A_0}{A_0} + \frac{18 + 4\log A_0}{A_0^2}. \tag{12}$$

For example, the coefficients of its linear constant and logarithm are bounded by $5167/240 < 22$ and $141/10 < 15$; its quadratic constant is < 18. At the smallest corner,

$$\Delta(2,3) = \alpha(11/16 + \ell/2) - 1/3 - 5\ell/12 > 2885/521272 > 1/200.$$

Indeed convexity bounds

$$\zeta(2) < \sum_{n=1}^{5} n^{-2} + \int_{11/2}^{\infty} x^{-2}dx = 65159/39600.$$

Insert $\alpha > 39600/65159$ first and $\ell < 7/10$ second. Elementary finite positive exponential or atanh series give

$$3/5 < \alpha < 2/3, \quad 2/3 < \ell < 7/10, \quad \log 3 < 11/10, \quad \log 5 < 17/10.$$

At $A_0 = 2^{16}$, $\log A_0 < 56/5$, and

$$\mathcal{E} < \frac{190}{65536} + \frac{314/5}{65536^2} < 1/300.$$

The positive error terms decrease thereafter. Hence every two-prime modulus $M \ge 2^{32}$ has $\mu(M) < -1/600$.

The sharper three cutoffs follow from the refined parameters, still without finite-height experiments:

$$\begin{aligned} &p=2,\ r\ge 11,\ M\ge 384^2: && \mu(M)<-1/600,\\ &p=3,\ r\ge 7,\ M\ge 512^2: && \mu(M)<-1/350,\\ &p\ge 5,\ r\ge 7,\ M\ge 600^2: && \mu(M)<-1/750. \end{aligned} \tag{13}$$

Here are the exact parameter substitutions and endpoint comparisons.

For $p=2$ the periodic extrema are $h=2u, m=-u$; they occur at $n=r-2, r+1$. Thus $K=4u, W=3u$. For $r\ge 11$, using the second coherent primitive and density choices,

$$u\le \tfrac{1}{2},\quad \kappa<\tfrac{4}{9},\quad \theta<\tfrac{1}{3},\quad b_0\le \tfrac{11}{4},\quad b_1\le \tfrac{3}{4},\quad b_0-b_1\le 2,\quad K\le 2,\quad E_0\le \tfrac{125}{44},\quad E_1\le \tfrac{3}{4}.$$

Substitution in (11), using $\beta<1/5$ and $1/\sqrt{2}<3/4$, gives

$$\mathcal{E}<\frac{363/20+(171/40)\log A_0}{A_0}+\frac{17/2+(3/2)\log A_0}{A_0^2}.$$

The unrounded linear constant is $143521/7920<363/20$. At $A_0=384$, $\log 384<6$, so

$$\mathcal{E}<73/640+35/294912<23/200.$$

The exact corner after inserting $\alpha>3/5$, then $\ell>2/3$, is

$$\Delta(2,11)>109/240-761/2178+(157/7260)(2/3)=3461/29040>7/60.$$

This proves the first line of (13).

For $p=3, r\ge 7$, (2) gives

$$u\le \tfrac{2}{3},\quad \kappa<\tfrac{1}{2},\quad \theta<\tfrac{1}{3},\quad K\le \tfrac{8}{3},\quad b_0\le \tfrac{14}{3},\quad b_1\le \tfrac{4}{3},\quad b_0-b_1\le \tfrac{10}{3},\quad E_0\le \tfrac{250}{63},\quad E_1\le \tfrac{10}{9}.$$

Therefore

$$\mathcal{E}<\frac{27+7\log A_0}{A_0}+\frac{13+(9/4)\log A_0}{A_0^2}.$$

The unrounded linear constant is $133843/5040<27$. Since $\log 512<63/10$,

$$\mathcal{E}(512)<711/5120+1087/10485760<7/50.$$

The corner computation is

$$\Delta(3,7)>603/1280-1307/3136+(1087/7840)(2/3)=27613/188160>1/7,$$

which proves the second line.

Finally, at $p=5, r=7$, insertion of $\alpha>3/5$, then $\ell>2/3$, gives $\Delta(5,7)>95869/470400>61/300$. The coefficient of ℓ after the first insertion is positive. Since $600=2^3\cdot 3\cdot 5^2$, $\log 600<33/5$, and (12) gives

$$\mathcal{E}(600)<121/600+37/300000=60537/300000<101/500.$$

The difference $61/300-101/500=1/750$ proves the third line. All endpoint comparisons are strict rational inequalities, and all error terms decrease above the stated scale.

Coverage boundary. The three cutoffs cover every pair except radicals $6, 10, 14, 15$. They reduce the remaining problem to a bounded domain, but do not themselves certify that domain or those four families. A complete proof of the all-modulus theorem must additionally supply their separate family arguments, the later source-domain reductions and the exact finite-certificate clauses. No sign extrapolation or monotonicity of the true finite means may replace these remaining premises.

B The refined radical-six balance

This argument applies to every M with $\mathrm{rad}(M) = 6$, not just 6^N. Put $a = \sqrt{M}$, $\alpha = \zeta(2)^{-1}$, $\ell = \log 2$, $\theta = \alpha/2$, and $\theta_6 = \alpha/12$. Period six gives

$$-2/3 \le U_6(x) - x/3 \le 2/3, \qquad -1/3 \le U_6(n) - n/3 \le 2/3 \quad (n \in \mathbb{Z}_{\ge 0}).$$

Partial summation yields $\sum_{d\le Q,(d,6)=1} 1/d \le 1 + \log Q/3$. For real $Q \ge 2$ the tail satisfies $Q\sum_{d>Q,(d,6)=1} d^{-2} \le 1/3 + 4/(3Q) \le 1$. For $1 \le Q < 2$, it is $< 2/9$ since $\sum_{d\ge 2,(d,6)=1} d^{-2} = 2\zeta(2)/3 - 1 < 1/9$. The real-cutoff estimate is therefore valid without rounding Q.

For $x = n + t$, $0 \le t < 1$, $|nU_6(n) - x^2/3| \le x$, by the displayed discrepancy. Möbius inversion with only six-unit contents consequently gives

$$|P(T) - \alpha T^2/2| \le (4/3)T + (1/3)T\log T. \tag{B.1}$$

Every closed interval of length h contains at most $h/3 + 4/3$ six-units, including boundary atoms. One proof uses the alternating spacings two and four: $2m + 1$ consecutive units span at least $6m$ and $2m$ units span at least $6m - 4$. For $L_T = (2T - M/T)_+$, the collision triangle in the preceding appendix then gives, when $2T^2 > M$,

$$J(T) \le L_T^2/18 + (8/9)L_T + 16/9. \tag{B.2}$$

At $2T^2 = M$, T is a nonunit, so the putative equality pair $d = f = T$ is forbidden. At $T^2 = M$, T is also a nonunit. Thus the same endpoint conventions apply for unequal valuations.

There is a useful additional gain in the far tail. Define

$$A(Q) = \sum_{\substack{d\le Q\\(d,6)>1}} \varphi(d)/d, \qquad A_6(Q) = \sum_{\substack{d\le Q\\6\mid d}} \varphi(d)/d.$$

Möbius expansion gives the exact identities

$$A(Q) = \sum_{\substack{d\le Q\\(d,6)=1}} \frac{\mu_{\mathrm{Mob}}(d)}{d}\left[\frac{1}{2}\lfloor Q/(2d)\rfloor + \frac{2}{3}\lfloor Q/(3d)\rfloor - \frac{5}{6}\lfloor Q/(6d)\rfloor\right],$$

$$A_6(Q) = \frac{1}{3}\sum_{\substack{d\le Q/6\\(d,6)=1}} \frac{\mu_{\mathrm{Mob}}(d)}{d}\lfloor Q/(6d)\rfloor.$$

The first bracket minus $Q/(3d)$ has period six and lies in $[-2/3, 1/3]$. The harmonic and real tail bounds above then give, for real $Q \ge 1$,

$$|A(Q) - \theta Q| \le E(Q) = 1 + (2/9)\log Q, \quad |A_6(Q) - \theta_6 Q| \le E_6(Q) = 2/3 + (1/9)\log Q. \tag{B.3}$$

For example, decompose a denominator into its powers of two and three and its six-unit part; summing the powers gives geometric series, and the six-unit harmonic and tail estimates above bound the remaining floor errors. Equivalently insert $p = 2, r = 3$ into the exact deleted-density floor identity in Appendix A, and separately retain denominators divisible by both primes. The resulting coefficients are $\theta = \alpha/2$ and $\theta_6 = \alpha/12$.

For centers divisible by six the neighbor repair costs d, rather than $2d$. With F defined in Section 4, the main survival envelope is therefore

$$V(t) = (\theta - \theta_6)F(t/\sqrt{2}) + 2\theta_6 F(t) + \theta/(2t^2), \quad t \ge \sqrt{2}.$$

In partial summation, the difference between the two rational kernels is unimodal; its maximum is at most $2M^2/T^3$ and its total variation is at most twice this. The extra error is consequently $8E_6(M/T)M/T^3$; the original one is $4E(M/T)/T$. Termwise positive-series integration, or the identity $2\,\mathrm{Li}_2(1/2) = \zeta(2) - \ell^2$, gives

$$\int_{\sqrt{2}}^{\infty} V(t)\frac{dt}{t} = 1/4 - 3\alpha/16 - \alpha(1-\ell)^2/24.$$

The dilogarithm identity follows by differentiating $\mathrm{Li}_2(x) + \mathrm{Li}_2(1-x) = \zeta(2) - \log x \log(1-x)$ and fixing the endpoint.

Repeat the central and middle integrals of the preceding appendix with (B.1)–(B.2), and start the far sum at $\left\lfloor \sqrt{2M} \right\rfloor + 1$. The complete result is

$$\mu(M) \le -\Delta_6 + \mathcal{E}_6(a),$$

where

$$\Delta_6 = \alpha(11/16 + \ell/2) - 1/3 - 5\ell/12 + \alpha(1-\ell)^2/24 > 1/130$$

and, writing $\beta = 3 - 2\sqrt{2}$,

$$\begin{aligned}\mathcal{E}_6(a) = \frac{1}{a}\big[&\alpha + 2 + (2/3)\log a + (8/9)\beta \\ &+ \ell(2\alpha + 32/9 + (2/3)\log a)/2 \\ &+ (1 + 5\alpha/8 + 4E(a) + (4/3)E_6(a))/\sqrt{2}\big] \\ &+ \{16\ell/9 + 2E(a) + 2E_6(a)\}/a^2.\end{aligned}$$

The margin follows from $\alpha > 39600/65159$, $\ell < 7/10$: the part without the final gain exceeds $2885/521272$, and the gain exceeds $1/440$, whose sum exceeds $1/130$. At $a_0 = 6^{9/2} > 3000$, $\log a_0 < 81/10$; using $\alpha < 2/3, \ell < 7/10, 1/\sqrt{2} < 3/4, \beta < 1/5$ gives

$$\mathcal{E}_6(a_0) < 23/a_0 + 10/a_0^2 < 6901/900000 < 1/130.$$

Every error term decreases for $a \ge a_0$. Hence $\mu(M) < -287/11700000$ whenever $\mathrm{rad}(M) = 6$, $M \ge 6^9$. The eight smaller diagonal moduli $6^1, \dots, 6^8$ are finite certificates; unequal valuations below the cutoff are handled separately.

C Elementary constant enclosures

All nonrational constants used for cutoff comparisons admit positive series enclosures. In particular

$$\sum_{n=1}^{10} n^{-2} + \frac{1}{11} > \frac{41}{25}, \qquad \zeta(2) < \sum_{n=1}^{5} n^{-2} + \frac{2}{11} = \frac{65159}{39600} < \frac{33}{20}.$$

For the upper bound, convexity of x^{-2} bounds the term at each integer $n \ge 6$ by its integral on $[n - 1/2, n + 1/2]$. The lower bound uses the integral tail after ten terms. Further,

$$e \le \sum_{n=0}^{6} \frac{1}{n!} + \frac{1}{7!(1 - 1/8)} < \frac{87}{32} < 3.$$

The atanh enclosure in Section 3, applied to rational arguments, proves all logarithmic comparisons with outward rounding. For instance its range reduction gives $2/3 < \log 2 < 7/10$, $\log 3 < 11/10$, $\log 5 < 17/10$, $\log 140 < 5$, $\log 384 < 6$, $\log 512 < 63/10$, $\log 600 < 13/2$. Alternatively, lower partial sums of $\exp x$ prove each upper logarithm comparison. These checks do not rely on floating decimal evaluation.

D The short-length bridges

Here we make explicit the predecessor result at lengths at least two used in Appendix E. This order avoids a circular use of the length-one completion. First the three sixth-power certificates for $B = 10, 12, 14$, the eight base-six certificates, and the coarse cutoffs give every two-prime base at $N \geq 6$, as explained in the family proof. The complete radical families then follow without using any length-two or length-one result.

Outside radicals $6, 10, 14, 15$, the coarse balance admits further corner cutoffs. Monotonicity of its leading margin gives

$$\Delta(2, 11) > 3461/29040 > 1/9,$$
$$\Delta(3, 7) > 27613/188160 > 1/7, \quad \Delta(2, 17) > 1447/10404 > 1/8.$$

At $a = 1296 = 6^4$, using $\log a < 36/5$, the coarse error $(22 + 15 \log a)/a + (18 + 4 \log a)/a^2$ is $< 46813/466560 < 1/9$. At $a = 1024$, using $\log a < 7$, it is $< 65047/524288 < 1/7$ and $< 1/8$. These comparisons and decreasing errors give:

remaining prime factors	sufficient M
$p = 2,\ r = 11$ or 13	6^8
$p = 2,\ r \geq 17$	2^{20}
$p \geq 3,\ r \geq 7$	2^{20}.

(D.1)

In particular all lengths $N \geq 5$ are covered. For length four the complete remaining bases are $21, 22, 26$; their three finite certificates prove all lengths $N \geq 4$.

For length three define the conservative domain of bases $B = p^a r^b$ outside the four complete radicals by $B^3 < 6^8$ if $p = 2, r = 11, 13$, and $B^3 < 2^{20}$ otherwise. Remove bases already covered by reindexing at length at least four. This gives exactly the following 35 bases:

$$21, 22, 26, 33, 34, 35, 38, 39, 44, 46, 51, 52, 55, 57, 58, 62, 63, 65, 68, 69,$$
$$74, 76, 77, 82, 85, 86, 87, 88, 91, 92, 93, 94, 95, 99, 104.$$

The retained domain generator also gives their unique prime valuations. Every one has a negative certificate, proving all lengths $N \geq 3$. The numerical limits behind the enumeration are $B \leq 118$ in the first category and $B \leq 101$ in the others.

For length two enumerate every integer base $B \leq 1295$ with exactly two prime factors outside the four radicals. Keep

$$B < 1296 \quad (p = 2, r = 11, 13), \qquad B < 1024 \quad \text{otherwise},$$

and remove $B = 441, 484, 676$, which are respectively the squares of the already certified bases $21, 22, 26$. Any other higher-power reindexing in this range is either in a complete radical family or above its analytic cutoff. There are exactly 454 retained bases, split into 256 odd bases, 183 bases with $p = 2, r \geq 17$, and 15 with $p = 2, r = 11, 13$. Their physical modulus is $M = B^2$; they are not 454 arbitrary two-prime moduli. The parameterized balance with the actual periodic discrepancy and outward constant enclosures pays 355 of them. The other 99 have negative histogram/log certificates constructed by Section 3. This proves every two-prime base at length two. All lengths $N \geq 2$ are now proved before the length-one residual inventory is formed.

The arithmetic in the analytic filters is also fully finite. For $\zeta(2)$ use

$$\sum_{j=1}^{K} j^{-2} + 1/(K+1) < \zeta(2) < \sum_{j=1}^{K} j^{-2} + 1/K,$$

invert outward to enclose α, and enclose logarithms by the atanh series. Square roots have rational enclosures obtained by integer square roots. The length-two filter uses $K = 128$ for the zeta sum; the companion records the root scale 10^{15}, logarithmic precision and any outward dyadic grid separately. A row is paid only if its entire upper enclosure is strictly negative; an inconclusive enclosure remains in the finite certificate set. The later length-one five-bound filter is evaluated in the same outward manner. A rounded decimal sign is never a filter.

Bridge	Rows	Kind of retained evidence
Odd prime powers	21	exact heights and integer Jensen sums
Base six, $N = 1, \dots, 8$	8	upper histograms and integer log sums
Length six, bases $10, 12, 14$	3	upper histograms and integer log sums
Squarefree semiprime prefix	4596	upper histograms and log ledger
Offdiagonal radical six	143	upper histograms and log ledger
Radical ten	97	upper histograms and log ledger
Radicals fourteen/fifteen	132	upper histograms and log ledger
Length four	3	upper histograms and log ledger
Length three	35	upper histograms and log ledger
Length two residual	99	upper histograms and log ledger
Final length one	4410	upper histograms and log ledger

Table 2: Finite proof dependencies. Domains are specified above and in the companion. Rows with different proof roles may overlap and are not to be added as a count of distinct moduli.

E Effective reduction and finite completion for two prime factors

Write

$$M = p^a q^b, \qquad p < q \text{ distinct primes}, \qquad a, b \geq 1, \qquad \mu(M) = \frac{1}{M} \sum_{s=0}^{M-1} \log R_M(s) - \frac{1}{2} \log M.$$

The minimum defining R_M uses positive denominators coprime to the whole M, and zero is one of the M equally weighted states. All logarithms in this appendix are natural. The reductions below apply to the actual valuations a, b; they do not compare means on different moduli. We write μ_{Mob} for the Möbius function, to distinguish it from the centered mean $\mu(M)$.

E.1 Exact discrepancy parameters and admissible errors

Put

$$D = pq, \quad u = \frac{(p-1)(q-1)}{pq}, \quad \rho = \frac{pq}{(p+1)(q+1)},$$
$$\alpha = \frac{1}{\zeta(2)}, \quad \ell = \log 2, \quad \kappa = \alpha\rho, \quad \theta = \alpha(1-\rho), \quad \beta = 3 - 2\sqrt{2}.$$

For integer $n \geq 0$, let

$$U_D(n) = n - \lfloor n/p \rfloor - \lfloor n/q \rfloor + \lfloor n/D \rfloor, \qquad e(n) = U_D(n) - un.$$

The function e is D-periodic and

$$e(n) - e(n-1) = \mathbf{1}_{(n,D)=1} - u, \qquad e(D-n) = \mathbf{1}_{(n,D)=1} - e(n) \quad (0 < n < D).$$

In particular,

$$-u \le e(n) \le 2u. \tag{14}$$

Indeed a minimum occurs at zero or a nonunit. For $n = pi < D$, write $pi = tq + r$, $1 \le r < q$, $0 \le t < p$. Then

$$e(pi) = \frac{r - i}{q}, \qquad i - r = tq - (p-1)i \le \frac{tq - (p-1)}{p} \le \frac{(p-1)(q-1)}{p} = uq.$$

The other nonunit family is treated with the primes interchanged. Reflection bounds a nonunit above by u. For a unit, its reflected unit follows a step of size $1 - u$ from a value at least $-u$; reflection then bounds the original value above by $2u$.

Let σ be the maximum of $e(n)$ over nonunits in $0 \le n < D$, including $n = 0$. The minimum over the whole period is $-\sigma$, and its maximum is $u + \sigma$: a maximum is immediately before a nonunit step, including the wraparound predecessor $D - 1$. An exact short formula, not a scan of all D residues, is

$$\begin{aligned} i_t &= \min\left(q - 1, \left\lfloor \frac{q(t+1) - 1}{p} \right\rfloor\right), && 0 \le t < p, \\ \sigma &= \max\left(0, \max_{0 \le t < p} \frac{(p-1)i_t - qt}{q}, \max_{1 \le j < p} \frac{(qj \bmod p) - j}{p}\right). \end{aligned} \tag{15}$$

To prove this, the nonunits are pi, $1 \le i < q$, and qj, $1 \le j < p$. On the block $t = \lfloor pi/q \rfloor$, the first numerator is increasing in i, so its maximum is at i_t. Every such block is nonempty since $q > p$; the minus one in (15) preserves its strict upper endpoint. There are at most $2p - 1$ nonzero candidates.

Define

$$K = 2u + 2\sigma, \qquad W = 2u + \sigma.$$

The real discrepancy lies between $-\sigma - u$ and $u + \sigma$. Consequently a closed interval of length h contains at most $uh + K$ positive D-units, with both boundary atoms included. Also

$$\sum_{\substack{d \le Q \\ (d,D)=1}} \frac{1}{d} \le u + (u + \sigma) + u \log Q \qquad (Q \ge 1).$$

For $x = n + t$, $n = \lfloor x \rfloor$, $0 \le t < 1$, the same discrepancy bounds give $\left|nU_D(n) - ux^2\right| \le Wx$. Möbius inversion and its integer-cutoff tail therefore yield

$$\left|P(T) - \kappa T^2\right| \le T(b_0 + b_1 \log T), \qquad P(T) = \sum_{\substack{d \le T \\ (d,D)=1}} \mu_{\mathrm{Mob}}(d) \lfloor T/d \rfloor U_D(\lfloor T/d \rfloor), \tag{16}$$

with either admissible pair

$$(b_0, b_1) = (3u + \sigma,\ 2u + \sigma), \tag{17}$$

$$(b_0, b_1) = ((2u + \sigma)^2 + u,\ u(2u + \sigma)). \tag{18}$$

The coefficient follows from

$$u \sum_{(d,D)=1} \frac{\mu_{\mathrm{Mob}}(d)}{d^2} = \frac{u\alpha}{(1 - p^{-2})(1 - q^{-2})} = \kappa.$$

For the first pair use $\sum_{d \le T} d^{-1} \le 1 + \log T$; for the second use the displayed unit harmonic bound. The omitted series costs at most uT. Thus $b_0 - b_1$ is respectively u and $(2u+\sigma)(u+\sigma)+u$, both nonnegative. The independent coarse choice $(b_0, b_1, K) = (4, 3, 4)$ is also valid.

For deleted denominators, set

$$A(Q)=\sum_{\substack{h\le Q\\(h,D)>1}}\frac{\varphi(h)}{h},\quad v=2-\frac{1}{p}-\frac{1}{q},\quad c_0=\frac{p-1}{p^2}+\frac{q-1}{q^2}-\frac{D-1}{D^2}.$$

Decomposing squarefree totient divisors gives the exact identity

$$A(Q)=\sum_{\substack{d\le Q\\(d,D)=1}}\frac{\mu_{\mathrm{Mob}}(d)}{d}\left[(1-1/p)\left\lfloor\frac{Q}{pd}\right\rfloor+(1-1/q)\left\lfloor\frac{Q}{qd}\right\rfloor-(1-1/D)\left\lfloor\frac{Q}{Dd}\right\rfloor\right].$$

The full sum's leading coefficient is θ. The bracket's linear coefficient is c_0Q/d, and its error is at most v; for a real cutoff, $Q\sum_{d>Q}d^{-2}<2$, not merely one. Hence either of the complete errors

$$E_{\mathrm{r}}(x)=v(2u+\sigma)+2c_0+vu\log x,\qquad E_{\mathrm{u}}(x)=3+2\log x \tag{19}$$

pays $|A(Q)-\theta Q|$ for real $Q\ge 1$. The refined constant also bounds the below-one partial-summation interval: $\theta<1/3$, whereas $vu\ge 7/18$. In particular,

$$\sum_{\substack{h\le Q\\(h,D)>1}}\varphi(h)\le\frac{\theta Q^2}{2}+2QE(Q).$$

E.2 Finite balance, closed endpoints and the four simultaneous gains

For every integer $T^2\le M$, the original central count satisfies

$$C_M(T)=1+2P(T)-J_M(T).$$

Reduction of common content is legitimate because it divides a full-unit denominator. Same-sign primitive images are injective. Opposite-sign images meet precisely at $Ad+Eb=M$, with positive $A,E,b,d\le T$. Thus $b+d\ge M/T$, with equality retained, and $(b,d)=1$. For a fixed ordered denominator pair the allowed phase interval has width at most $[T(b+d)-M]/(bd)\le 1$; equality would require $M=T^2$ and a denominator T, which is nonunit at a square modulus. Putting $L_T=(2T-M/T)_+$, the closed interval unit bound gives

$$J_M(T)\le\frac{u^2L_T^2}{2}+2uKL_T+K^2\quad(2T^2>M).$$

If $2T^2=M$, equality would force $b=d=T$; the odd prime divisor of M divides T, so the collision is absent. Thus $J_M(T)=0$ also at that boundary.

Here is the far-cover argument used in the finite balance. For $T^2>2M$, full Farey neighbors of order T have coprime denominators and sum greater than T. Two retained neighbors cover their intervening arc. If both are deleted, their prime obstructions are complementary, so their denominator difference is a full unit; when both denominators are at least M/T, that difference witness covers the arc. An uncovered point must therefore touch a deleted center c/h with $h<M/T<T/2$. Its complete neighbors have denominators greater than $T-h$; subtracting c/h once repairs a bad neighbor and gives a determinant-one unit neighbor of denominator greater than $T-2h$. When $D\mid h$, the complete neighbors were already units. The corresponding open hole has length at most

$$\frac{2(M-hT)}{h(T-j_hh)},\qquad j_h=\begin{cases}1, & D\mid h,\\ 2, & D\nmid h.\end{cases}$$

One additional integer per reduced center pays rounding; overlaps only enlarge the upper bound. Therefore

$$M - C_M(T) \le \sum_{\substack{2\le h<M/T\\(h,D)>1}} \varphi(h) + \sum_{\substack{2\le h<M/T\\(h,D)>1}} \frac{2\varphi(h)(M-hT)}{h(T-j_h h)}. \tag{20}$$

The denominators are strictly positive. No center phase or exact tail distribution is needed.

Write $x=\sqrt{M}$, $k=\lfloor x\rfloor$, $t=x-k$. For any compatible primitive pair and deleted error E, define

$$\Delta = \rho\{\alpha(5/8+\ell)+1/2\} + 3\alpha/8 - 1/2 - \ell/2 - \tfrac{3}{4}(1-\ell)u^2, \tag{21}$$

$$\begin{aligned}\mathcal{E}_{b,E}(x) = {}& \frac{2\kappa + 2(b_0-b_1) + 2b_1\log x + 2uK\beta}{x}\\ &+ \frac{\ell}{2x}(4\kappa + 2b_0 + 2b_1\log x + 2uK)\\ &+ \frac{1+5\theta/4+4E(x)}{\sqrt{2}x} + \frac{K^2\ell + 2E(x)}{M}.\end{aligned} \tag{22}$$

$$\delta_x = \frac{1+2t-3t^2}{2} + \frac{t^3}{x}, \qquad G_x = 2x(1-t)+t^2 = 2x - M + k^2,$$

$$\mathcal{G}_b(x) = \frac{\ell\kappa G_x + 2\kappa\delta_x + 2(b_0-b_1) + \log x}{M}. \tag{23}$$

Then the strengthened finite balance is

$$\mu(M) \le -\Delta + \mathcal{E}_{b,E}(x) - \mathcal{G}_b(x). \tag{24}$$

For completeness, its integration and rounding costs can be checked directly. The centered mean is minus the central CDF integral up to x, plus the survival integrals over $[x,\sqrt{2}x]$ and $[\sqrt{2}x,\infty)$. The collision integrals are

$$\frac{1}{M}\int_{x/\sqrt{2}}^{x}(2y-M/y)^2\frac{dy}{y} = \frac{3}{2} - 2\ell, \qquad \int_{x/\sqrt{2}}^{x}(2y-M/y)\frac{dy}{y} = x\beta.$$

The central credit and monotone middle plateau, before retaining gains, are respectively at least

$$\kappa - u^2(3/4-\ell) - \frac{2\kappa + 2(b_0-b_1) + 2b_1\log x + 2uK\beta}{x} - \frac{K^2\ell}{2M}$$

and at most

$$(1 - 2\kappa + u^2/2)\frac{\ell}{2} + \frac{\ell(4\kappa + 2b_0 + 2b_1\log x + 2uK)}{2x} + \frac{K^2\ell}{2M}.$$

For the far sum use $h < M/T$, partial summation of A, and

$$F(z) = \sum_{n\ge 1}\frac{z^{-2n}}{n(n+1)}, \qquad F(1) = 1.$$

The leading normalized survival is bounded by $\theta F(T/(\sqrt{2}x)) + \theta x^2/(2T^2)$. Its logarithmic integral is $(1-\rho)/2 - 3\theta/8$. The first integer far threshold is $\lfloor\sqrt{2M}\rfloor + 1$; the initial interval costs at most $1/(\sqrt{2}x)$, including equality at an integer $\sqrt{2M}$. The first summand and discrete error add $(5\theta/4 + 4E(x))/(\sqrt{2}x) + 2E(x)/M$. These terms give (21)–(22).

The four gains in (23) affect distinct terms. Indeed

$$\int_1^x \lfloor y\rfloor^2\frac{dy}{y} \ge \frac{(k-1)^2}{2} + \frac{k^2 t}{x} = \frac{x^2}{2} - x + \delta_x.$$

Here $\log(1+1/j) \ge 1/j - 1/(2j^2)$ pays the complete intervals and $\log(x/k) \ge t/x$ pays the last interval. Exact primitive-error integration restores $2(b_0-b_1)/M$; the isolated zero restores $(\log x)/M$. Retaining the exact endpoint k^2, instead of $M-2x$, restores $\ell\kappa G_x/M$ on the middle plateau. Collision and far penalties are unchanged, so all four improvements coexist. At a square, $\delta_x = 1/2$, $G_x = 2x$. They must remain paired with their own b_0, b_1.

For the coarse pair $(4,3,4)$, the error in (22) is strictly below

$$\mathcal{E}_{\mathrm{c}}(x) = \frac{22+15\log x}{x} + \frac{18+4\log x}{M}.$$

This follows from $2\kappa \le 4/3$, $\theta \le 1/3$, $u \le 1$, $\ell < 7/10$, $\beta < 1/5$, $1/\sqrt{2} < 3/4$: the linear constant is at most $5167/240 < 22$, its logarithmic coefficient is at most $141/10 < 15$, and the quadratic constant is less than 18. Thus the following are five separately valid, complete upper bounds:

$$-\Delta + \mathcal{E}_{\mathrm{c}} - \mathcal{G}_{(4,3)}, \qquad -\Delta + \mathcal{E}_{b,E} - \mathcal{G}_b, \quad b \in \{(17),(18)\},\ E \in \{E_{\mathrm{r}}, E_{\mathrm{u}}\}.$$

Taking their minimum is legitimate; moving a gain from one bound to another is not.

E.3 Uniform modulus cutoffs

The leading credit increases with either prime parameter. Put $z = 1/p, w = 1/q$, $A_0 = \alpha(5/8+\ell)+1/2$, $C_0 = 3(1-\ell)/4$. Multiplying its z-derivative by $(1+z)/\rho > 0$ gives

$$-A_0 + 2C_0(1-z)(1+z)^2(1-w)^2(1+w) < 0.$$

Indeed the two factors are at most $9/4$ and 1, respectively, $2C_0 < 1/2$, and $A_0 > 51/40 > 9/8$, using $\alpha > 3/5$ and $2/3 < \ell < 7/10$. This is monotonicity of the bound, not of the true mean.

Since $\sigma \le u$, the second primitive pair has

$$b_0 \le 9u^2+u, \quad b_1 \le 3u^2, \quad b_0 - b_1 \le 6u^2+u, \quad K \le 4u.$$

For $p=2$, reflection gives the exact values $\sigma = u$, $K = 4u$, $W = 3u$. Using the refined deleted error in (22), elementary substitution gives the following bounds and rational endpoint witnesses:

prime range	error upper	x_0	leading credit lower
$p=2,\ q\ge 11$	$\dfrac{363/20+(171/40)\log x}{x} + \dfrac{17/2+(3/2)\log x}{x^2}$	384	$\dfrac{3461}{29040} > \dfrac{7}{60}$
$p=3,\ q\ge 7$	$\dfrac{27+7\log x}{x} + \dfrac{13+(9/4)\log x}{x^2}$	512	$\dfrac{27613}{188160} > \dfrac{1}{7}$
$p\ge 5,\ q\ge 7$	$\mathcal{E}_{\mathrm{c}}(x)$	600	$\dfrac{95869}{470400} > \dfrac{61}{300}$

For the first two rows use respectively

$$c_0 \le 13/44,\ E_{\mathrm{r},0} \le 125/44,\ E_{\mathrm{r},1} \le 3/4, \quad c_0 \le 20/63,\ E_{\mathrm{r},0} \le 250/63,\ E_{\mathrm{r},1} \le 10/9.$$

For the second row $u \le 2/3$, $\kappa < 1/2$, $K \le 8/3$, $b_0 \le 14/3$, $b_1 \le 4/3$, $b_0 - b_1 \le 10/3$. The first row uses $u \le 1/2$, $\kappa < 4/9$, $K \le 2$, $b_0 \le 11/4$, $b_1 \le 3/4$, $b_0 - b_1 \le 2$. The resulting unrounded linear constants are $143521/7920 < 363/20$ and $133843/5040 < 27$. At the three endpoints,

$$\begin{aligned}
\mathcal{E}(384) &< 73/640 + 35/294912 < 23/200,\\
\mathcal{E}(512) &< 711/5120 + 1087/10485760 < 7/50,\\
\mathcal{E}_{\mathrm{c}}(600) &< 60537/300000 < 101/500.
\end{aligned}$$

Use $\log 384 < 6$, $\log 512 < 63/10$, $\log 600 < 33/5$; these follow from the positive exponential sums giving $\log 2 < 7/10, \log 3 < 11/10, \log 5 < 17/10$. The constant and logarithmic error terms decrease thereafter. It follows, for arbitrary positive valuations, that

$$\begin{aligned} &p = 2,\ q \geq 11,\ M \geq 147456: && \mu(M) < -1/600, \\ &p = 3,\ q \geq 7,\ M \geq 262144: && \mu(M) < -1/350, \\ &p \geq 5,\ q \geq 7,\ M \geq 360000: && \mu(M) < -1/750. \end{aligned} \tag{25}$$

Equality at these sufficient cutoffs is already covered.

E.4 The larger prime to the first power

Suppose $M = Kq$, $K = p^a$, $q > K$, $M \geq 19600$. In the single-p-mark model, all central denominators are smaller than q and are therefore also full units. Primitive cancellation and the collision-free range through $\lfloor \sqrt{M/2} \rfloor$ transfer unchanged from the single-mark proof. In its Farey tail, adjacent denominators cannot both be multiples of p; its deleted center bracket has denominators greater than $T - h$. Thus the odd-prime estimates of Section 4, and the binary calculation given in the following squarefree subsection, transfer to this modulus without a prime-power assumption:

p	leading credit	error at $x = \sqrt{M}$
2	1/12	$(33/8 + (12/11)\log x)/x + 1/(4x^2)$
3	41/486	$(107/27 + (16/11)\log x)/x + 1/(3x^2)$
≥ 5	91/800	$(151/40 + (24/11)\log x)/x + 1/(2x^2)$

At $x = 140$, $\log 140 < 5$ bounds the errors by quantities strictly below $7/100, 82/1000, 106/1000$. Explicit upper fractions are

$$59021/862400, \quad 467279/5821200, \quad 22619/215600.$$

All three errors decrease for $x \geq 140$. Their strict single-mark margins are respectively greater than $1/75, 287/121500, 31/4000$.

Below real height q, single-mark and full-unit events coincide. For a complete reduced lattice basis, the two-prime short-basis argument gives $R_M \leq \sqrt{2}M/\lambda_1$. The all-state incidence bound $\Pr(\lambda_1 \leq Y) \leq 3Y^2/M$ therefore implies $\Pr(R_M > X) \leq \min(1, 6M/X^2)$. If $q \geq 6K$, the extra full-unit logarithmic cost is at most

$$\int_q^\infty \frac{6M}{X^2}\frac{dX}{X} = \frac{3K}{q}.$$

Consequently

$$b = 1, \quad M \geq 19600, \quad q \geq c(p)p^a \quad \Longrightarrow \quad \mu(M) < 0, \qquad c(p) = \begin{cases} 225, & p = 2, \\ 1300, & p = 3, \\ 400, & p \geq 5. \end{cases} \tag{26}$$

These ratios imply $q > K$ and $q \geq 6K$. The comparisons are $3/225 = 1/75$, $287/121500 - 3/1300 = 43/789750 > 0$, and $31/4000 - 3/400 = 1/4000 > 0$. The first equality still pays a strict result because its single-mark margin is strict. The proof has $b = 1$; it is not a radical-only rule and cannot be applied with the prime roles reversed.

E.5 Squarefree semiprimes: zero mark loss and the complete prefix

Let $R_{p;M}$ denote the minimum with only $p \nmid d$. Separating witnesses according to whether $q \mid d$ gives exactly

$$R_{p;M}(s) = \min\{R_M(s),\ qR_{p;M/q}(s \bmod M/q)\}.$$

If $q \mid d$, the congruence forces $q \mid A$, and division by q gives the displayed child congruence; conversely multiplying a child witness by q gives a sole-p witness. No arbitrary nonunit content is cancelled. At $M = pq$, the circle-gap cap proved below is $R_{pq} \le q$, while the second term is at least q. Consequently $R_{pq} = R_{p;pq}$ pointwise. Since $\sqrt{pq} < q$, the single-mark transfer of the preceding subsection applies, with NO mark-loss penalty. Its odd-prime estimates give $\mu(pq) < -1/500$ for $p \ge 3$, $pq \ge 19600$.

For the even exterior we spell out the smaller sufficient threshold. The binary single-mark calculation below is valid more generally for $M = 2^a q$, $q > 2^a$, with $x = \sqrt{M} < q$: all its central denominators are full units, and the sole-two-mark Farey tail depends only on the excluded even denominators. In particular it applies at $M = 2q$. The divisor union bound on pairs with odd denominator gives

$$P_2(T) \ge [1 - 3\zeta(2)/8]T^2 - \tfrac{1}{2}T\log T \ge \tfrac{3}{8}T^2 - \tfrac{1}{2}T\log T.$$

Indeed there are at least $T^2/2$ raw pairs; remove every odd content $h \ge 3$, bounding its count by $T^2/(2h^2) + T/(2h)$. The sums are bounded by $\sum_{h\ge3,\ h\ \mathrm{odd}} h^{-2} = 3\zeta(2)/4 - 1$ and $\sum_{3\le h\le T,\ h\ \mathrm{odd}} h^{-1} \le \log T$. Here $\zeta(2) < 5/3$. For $T \le K = \lfloor x/\sqrt{2}\rfloor$, $x = \sqrt{M}$, there are no collisions, and hence $C_M(T) \ge cT^2 - dT\log T$ with $c = 3/4, d = 1$. The quadratic floor integral and monotone plateau give

$$\frac{1}{M}\int_1^x C_M(\lfloor y\rfloor)\frac{dy}{y} \ge c\frac{K^2}{x^2}\left(\frac{1}{2} + \log\frac{x}{K}\right) - \frac{2cK}{x^2} - d\frac{K\log K}{x^2}\left(1 + \log\frac{x}{K}\right).$$

This follows by integrating to K, using $\int_1^K \lfloor y\rfloor^2 dy/y \ge K^2/2 - K$, bounding the primitive-error integral by $K\log K$, and retaining the plateau $C_M(K)\log(x/K)$. The displayed $2cK$ is a safe enlargement of the floor loss. For $x \ge 115$,

$$K/x > 7/10 - 1/115 > 11/16, \quad K/x < 3/4,$$
$$K^2/x^2 \ge 1/2 - \sqrt{2}/x, \quad \ell/2 \le \log(x/K) \le 5/11.$$

With $2/3 < \ell < 1$ and $\sqrt{2} < 3/2$, this yields central credit $5/16 - [9/4 + (12/11)\log x]/x$. The single-mark Farey tail has majorant

$$V(z) = \tfrac{1}{2}F(z) + \frac{1}{8z^2}, \quad z \ge 1, \qquad V(1) = 5/8, \qquad \int_1^\infty V(z)\frac{dz}{z} = \frac{\zeta(2)-1}{4} + \frac{1}{16} < \frac{11}{48}.$$

The initial interval costs $1/x$, the first summand $5/(8x)$, and the integer-hole rounding costs $1/(4x) + 1/(4x^2)$. Subtracting the central credit gives, for every $x \ge 115$,

$$\mu(2q) \le -\frac{1}{12} + \frac{33/8 + (12/11)\log x}{x} + \frac{1}{4x^2}.$$

The error decreases there. Since $\log 115 < 5$, its remaining strict margin at $x = 115$ is

$$\frac{1}{12} - \frac{843}{10120} - \frac{1}{52900} = \frac{1127}{80302200} > 0.$$

Thus every odd prime $q \ge 6613$ lies in this analytic exterior: $2q \ge 13226 > 115^2$.

The complete prefix is exactly

$$\{2q : 3 \le q < 6613,\ q \text{ prime}\} \ \cup\ \{pq : 3 \le p < q,\ p, q \text{ prime},\ pq < 19600\}.$$

It has 853 even and 3743 odd moduli, 4596 in total. Every row has a negative outward-log certificate obtained from a legal full-unit upper histogram, capped at the proved q; the pointwise construction is proved in Section 3.2, and the logarithmic certificate inequality is given below. This finite prefix plus the analytic exterior proves every squarefree-semiprime mean strictly negative, without any transfer to higher valuations or any sampling argument.

E.6 Complete small-radical families and their finite witnesses

For clarity, the earlier complete families have an explicit infinite part and a separately retained finite part. The uniform coarse balance gives the cutoffs 2^{22} for odd prime pairs and $9\,000\,000$ for $p = 2, q \ge 5$: at $x = 2048$ its error is less than $7/100$ and $\Delta(3, 5) > 1691/14400 > 1/10$; at $x = 3000$ its error is less than $1/20 - 2026/937500$ and $\Delta(2, 5) > 263/4500 > 1/20$.

The separately proved refined radical-six finite balance gives $\mu(M) < -287/11700000$ for every modulus of radical six with $M \ge 6^9$. It retains the shorter repaired bracket at centers divisible by both primes, and applies to unequal valuations, with nonunit central equality endpoints and a strictly started far sum. This preceding radical-six lemma, rather than an asymptotic sign or the coarse error above, is used for that one infinite family.

The following finite witness table completes the families. Each entry means that every original modulus in the specified set has a retained negative outward-log upper certificate, not that its true minimum distribution was fully enumerated.

family	finite set below its analytic cutoff	rows
6^n	$1 \le n \le 8$	8
2^a3^b	$M < 6^9,\ a \ne b,\ \gcd(a, b) < 6$	143
2^a5^b	$M < 9\,000\,000,\ (a, b) \ne (1, 1),\ \gcd(a, b) < 6$	97
2^a7^b	$M < 9\,000\,000,\ (a, b) \ne (1, 1),\ \gcd(a, b) < 6$	78
3^a5^b	$M < 2^{22},\ (a, b) \ne (1, 1),\ \gcd(a, b) < 6$	54
B^6	$B \in \{10, 12, 14\}$	3

These rows are not summed into a single disjoint inventory: their roles in the dependency proof differ. The first row plus the six-radical cutoff proves the base-six all-length statement. Every two-prime base at length at least six is then covered: at length six, bases at least fifteen have modulus greater than 6^9, base six is already covered, and the other bases below fifteen are exactly $10, 12, 14$, the last row. For lengths at least seven, base ten reaches its even cutoff, and bases at least twelve reach 6^9. Thus a modulus with $\gcd(a, b) \ge 6$ is covered by reindexing at that same modulus. The middle four rows, the squarefree-semiprime theorem, and their respective large-modulus cutoffs now exhaust every valuation in radicals $6, 10, 14, 15$.

E.7 Certificate arithmetic and exhaustive final partition

The finite certificates use disjoint original gcd strata. For $g \mid M$, $g < M$, put $m = M/g$ and

$$P_g(T) = \sum_{\substack{e \le T/g \\ (e,D)=1}} \mu_{\mathrm{Mob}}(e) U_m(\lfloor T/(ge) \rfloor) U_D(\lfloor T/e \rfloor).$$

Through $\lfloor \sqrt{M} \rfloor$, the exact stratum CDF is $2P_g(T) - J_g(T)$, where ordered closed collision phases $Af + Ed = M$ are assigned to $g = \gcd(A, M)$. A centered inverse modulo m, of magnitude $h \le H = \lfloor (m-1)/2 \rfloor$, supplies numerator $\pm g$. If one prime v is missing from m,

repair a denominator divisible by v to $m-h$, reversing the numerator sign. The repair remains a unit of the original M. For $T \geq g$ its CDF is $2U_D(\min(T,H))$, with the extra pure-child term

$$2\max\{0, V_v(H) - V_v(\max(0, m-T-1))\}, \qquad V_v(z) = \lfloor z/v \rfloor - \lfloor z/D \rfloor.$$

Denominator one independently contributes $2U_m(\min(\lfloor T/g \rfloor, H))$. At $m=2$ each distribution is one atom at height g, not a doubled pair; both are zero for $T < g$. The circle-gap argument with p multiples of a state gives the global cap M/p: the index difference is less than p, hence full-unit, and its centered residue is at most M/p. The state M/p attains this cap. Thus each stratum is completely covered at

$$K_g = \min\{M/p,\ g\lfloor m/2 \rfloor,\ \max(g, D_m)\}.$$

$$D_m = \begin{cases} 1, & m = 2, \\ H, & \mathrm{rad}(m) = D, \\ \max(H, m-v), & m \text{ is a pure-prime power missing } v. \end{cases}$$

The earliest radical-six stratum batch used the wider cap $\lfloor m r_m/2 \rfloor$, with $r_m = 1$ for a mixed child and $r_m = v$ for a pure child missing prime v. This is also safe: in the mixed case it is at least H; in the pure case, $v \geq 2$ makes it at least m, hence at least every repaired denominator $m-h$. The singleton $m=2$ remains one explicit atom at height g. Later batches use the refined D_m above. These safe recipes are distinguished in the provenance map.

Take the maximum of these credits inside a stratum, sum only over disjoint strata, add zero once, and combine by maximum with the global far credit from (20). On an ordered calendar containing all integers through $\lfloor \sqrt{M} \rfloor$, successive $\lceil 257T/256 \rceil$ thereafter, all divisors/caps, the first strict far threshold, and M/p, cumulative maxima give integer lower CDF values L_i ending at M. The histogram $h_i = L_i - L_{i-1}$ dominates the true height even between knots, by CDF monotonicity. Therefore

$$\mu(M) \leq \frac{2\sum_i h_i u(t_i) - M l(M)}{2M\,2^{64}} = \frac{Q_M}{2M\,2^{64}}, \quad l(n) \leq 2^{64}\log n \leq u(n).$$

This doubled denominator is unchanged for odd M. The enclosures follow from binary range reduction and

$$0 \leq \log y - 2\sum_{j=0}^{K-1} \frac{z^{2j+1}}{2j+1} \leq \frac{2z^{2K+1}}{(2K+1)(1-z^2)}, \qquad z = \frac{y-1}{y+1}, \quad 1 \leq y \leq 2.$$

The finite assertion is $Q_M < 0$ on every row of each declared set. Large intermediate central counts rely on the exact formula and its generator; a retained checker is not described as a complete second implementation of every central count or of the true far survival.

We now use the previously proved squarefree-semiprime theorem and the two-prime all-length theorem for lengths at least two. If $e = \gcd(a,b) \geq 2$, set $C = p^{a/e}q^{b/e}$, so $M = C^e$; this is precisely the same state space and witness set. Together with the four complete radicals and (25), the only conservative residual is

$$\mathcal{R} = \{(M,p,q,a,b) :\ \gcd(a,b) = 1,\ (a,b) \neq (1,1),\ pq \notin \{6,10,14,15\},\ M < C(p)\}.$$

$$C(p) = \begin{cases} 147456, & p = 2, \\ 262144, & p = 3, \\ 360000, & p \geq 5. \end{cases}$$

It contains all positive singleton and unequal valuations not paid by another stated rule. Unique factorization gives one ordered tuple per physical modulus. Since $M \geq p^2 q > p^3$, it is sufficient to enumerate primes $p \leq 71$, primes $q < C(p)/p^2$, and positive prime powers subject to the exact strict product test and these predicates. A second enumeration by factorization gives the same full list.

Inside this residual the ratio rule (26) has only the following possible supports:

(p, a, b)	prime q range
$(2, 2, 1)$	$4900 \leq q < 36864$
$(2, 3, 1)$	$2450 \leq q < 18432$
$(2, 4, 1)$	$3600 \leq q < 9216$
$(3, 2, 1)$	$11700 \leq q < 262144/9$
$(5, 2, 1)$	$10000 \leq q < 14400$

Indeed $q \geq c(p)p^a$ and $p^a q < C(p)$ imply $c(p)p^{2a} < C(p)$, while $a = 1, b = 1$ is excluded. The lower endpoints also impose $M \geq 19600$. This proves that the rule excludes no unseen exponent orientation.

The exact coverage table is

stage	rows	mathematical reason
conservative residual $\mathcal{R}$	19239	complete original predicate
ratio-paid part	7857	(26)
after the ratio filter	11382	$19239 - 7857$
analytically certified part	6972	negative upper from the five complete bounds
finite remainder	4410	$11382 - 6972$
both primes at least five	1087	every retained $Q_M < 0$
smaller prime three	1422	every retained $Q_M < 0$
smaller prime two	1901	every retained $Q_M < 0$
remaining two-prime obligations	0	$1087 + 1422 + 1901 = 4410$

The 6972 analytic rows are a conservative certificate partition, not a claim that all other exact analytic bounds or true means are nonnegative. Each row retains all five bound enclosures. For α, the rational tails

$$\sum_{j=1}^{128} j^{-2} + \frac{1}{129} < \zeta(2) < \sum_{j=1}^{128} j^{-2} + \frac{1}{128}$$

are inverted outward. Integer square-root floors give rational $x_L \leq \sqrt{M} \leq x_U$ with the same fixed k and $t_U < 1$. The gains use the lower bounds

$$\delta_L = \min_{z \in \{t_L, t_U\}} \frac{1 + 2z - 3z^2}{2} + \frac{t_L^3}{x_U}, \qquad G_L = \max(1,\, 2x_L - M + k^2).$$

The first uses concavity, the second the exact identity $G_x = 1 + (1 - t)(2k - 1 + t) \geq 1$. Their denominator is the exact M, never an approximate root squared. After lowering α, write

$$\Delta_L = C_L + s_L \ell_\star, \quad C_L = \alpha_L(5\rho/8 + 3/8) + \rho/2 - 1/2 - 3u^2/4, \quad s_L = \alpha_L \rho - 1/2 + 3u^2/4,$$

where $\ell_\star = \ell_L$ for $s_L \geq 0$ and ℓ_U otherwise. Positive error coefficients are enclosed upward, while their own nonnegative gains are enclosed downward. Strictly negative resulting uppers pay the analytic rows. No sign-selected domain or interpolation is used.

Every original two-prime modulus is thus covered by a preliminary theorem, an explicitly proved reduction, or one of the complete finite certificate sets. We conclude $\mu(p^a q^b) < 0$ for all

distinct primes and positive valuations. For $B = p^\gamma q^\delta$, $\gamma, \delta \geq 1$, $N \geq 1$, put $M = B^N$. The original height, state measure and full-unit marks are identical, and $\log M = N \log B$. Division by $\log B > 0$ therefore gives

$$\mathbb{E} \log_B R_{(B,N)} < N/2.$$

This conclusion is restricted to exactly two prime divisors. It neither transfers to bases with three prime factors nor asserts monotonicity, optimal margins, exact true means or novelty.

F Primitive incidence distribution bounds

We supply the infinite argument and its finite complement for (S1). For $1 \leq Y < M$ a shortest lattice vector is primitive in lattice coordinates. Writing $(a, d) = (Mm + sd, d)$, this means $\gcd(m, d) = 1$, and does not mean $\gcd(a, d) = 1$. There is no horizontal nonzero vector in that disk; choose the sign $d > 0$. For a fixed $d \leq Y$, put $h = \gcd(d, M)$, $r = d/h$, $A = \sqrt{Y^2 - d^2}$, and $U_r(X) = \#\{b \in \mathbb{Z} : |b| \leq X,\ \gcd(b, r) = 1\}$. The exact number of primitive incidences with that denominator is

$$c_d(M, Y) = \frac{\varphi(d)}{\varphi(r)} U_r(A/h). \tag{A1}$$

Indeed $a = hb$ is necessary. In $b = rs + (M/h)m$, $\gcd(r, M/h) = 1$. The h possible states give all lifts of a residue of m mod r to m mod d. A lift is a unit exactly when that residue is a unit, and each unit residue has $\varphi(d)/\varphi(r)$ unit lifts. This also counts numerator zero, precisely for $r = 1$.

Complete periods and pairing residues $u, r - u$ give $U_r(X) \leq \varphi(r)(2X/r + 1)$. Thus, with $f_Y(t) = 2\sqrt{Y^2 - t^2} + t$,

$$I_{\mathrm{prim}}(M, Y) \leq \sum_{d=1}^{\lfloor Y \rfloor} \frac{\varphi(d)}{d} f_Y(d). \tag{A2}$$

The union bound divides this count by M.

For the 3/2 bound, majorize $\varphi(d)/d$ by the period-six weights $(1, 1/2, 2/3, 1/2, 1, 1/3)$. Their mean is $2/3$, and their cumulative discrepancy from that mean lies in $[0, 1/3]$ for all integer indices by periodicity. The function f_Y increases to $\sqrt{5}Y$ and then decreases. Abel summation bounds the discrepancy contribution by $\sqrt{5}Y/3$. A right Riemann sum, with its positive variation paid, gives

$$\sum_{d=1}^{\lfloor Y \rfloor} f_Y(d) \leq \frac{\pi + 1}{2} Y^2 + (\sqrt{5} - 2)Y.$$

Using $\pi < 22/7$ and $\sqrt{5} < 9/4$,

$$I_{\mathrm{prim}}(M, Y) < \frac{29}{21} Y^2 + \frac{11}{12} Y \leq \left(\frac{3}{2} - \frac{1}{224}\right) Y^2 \quad (Y \geq 8). \tag{A3}$$

The inequality $\pi < 22/7$ follows, for example, from the positive integral $\int_0^1 t^4(1-t)^4/(1+t^2)\, dt = 22/7 - \pi$.

For $1 \leq Y < 8$, squared norms are integers. Put $T = \lfloor Y^2 \rfloor$. An exact bound independent of M is

$$E(T) = \sum_{d=1}^{\lfloor \sqrt{T} \rfloor} \max_{h \mid d} \frac{\varphi(d)}{\varphi(d/h)} U_{d/h} \left(\frac{\lfloor \sqrt{T - d^2} \rfloor}{h} \right).$$

The complete finite table, for successive T, is

1–9	$1, 3, 3, 4, 7, 7, 7, 8, 10$
10–18	$12, 12, 12, 15, 15, 15, 17, 19, 21$
19–27	$21, 24, 24, 24, 24, 24, 28, 30, 30$
28–36	$30, 31, 31, 31, 33, 33, 37, 37, 39$
37–45	$41, 41, 41, 42, 44, 44, 44, 44, 48$
46–54	$48, 48, 48, 54, 60, 60, 62, 63, 63$
55–63	$63, 63, 63, 63, 63, 63, 63, 63, 63.$

Every entry satisfies $2E(T) \le 3T$; each is reproducible from divisors, integer square roots and gcd tests. Maximizing separately for each d only enlarges the count; no common maximizing modulus is postulated. This proves the $3/2$ bound. The remaining ranges $Y < 1$ and $Y \ge M$ are respectively empty and bounded trivially by one, including $M = 1$.

For the second bound let $P = 30030$ and

$$w_d = \prod_{p \mid P} \left(1 - \frac{1}{p}\mathbf{1}_{p \mid d}\right), \qquad \eta = \prod_{p \mid P}(1 - p^{-2}) = \frac{442368}{715715}.$$

These weights majorize $\varphi(d)/d$; their primes are an arithmetic device, not additional marks on denominators. Over the full period the discrepancy $D(n) = \sum_{d \le n}(w_d - \eta)$ has extrema

$$\ell = -134597/1431430, \quad h = 744773/1431430$$

(first occurring at 11376 and 18653). The finite predicate consists of all 30030 weights and all cumulative sums. Periodicity proves these bounds for every integer $n \ge 0$.

Concavity of $2\sqrt{Y^2 - t^2}$ and the trapezoid rule imply

$$\sum_{d=1}^{\lfloor Y \rfloor} f_Y(d) \le \frac{\pi + 1}{2} Y^2 - \frac{Y}{2} + \frac{1}{2}.$$

For clarity, writing $n = \lfloor Y \rfloor$, the semicircle sum is at most $\pi Y^2/2 - Y + \sqrt{Y^2 - n^2}$; use $\sqrt{u} \le (u+1)/2$ and add $n(n+1)/2$. Abel summation, using $D \ge \ell$ on the increasing part and $D \le h$ on the decreasing part, gives the discrepancy bound $[h\sqrt{5} - \ell(\sqrt{5} - 2)]Y$. The terminal term is included. Hence

$$I_{\text{prim}} \le \frac{\eta(\pi + 1)}{2} Y^2 + [-\eta/2 + (h - \ell)\sqrt{5} + 2\ell]Y + \eta/2.$$

Substitute $\pi < 22/7$, $\sqrt{5} < 161/72$, and divide by M: the coefficients are exactly a, b, d in (S1). They are nonnegative and $a > 1$, so the empty and trivial outer ranges are again covered. This proves (S1) for all M, x.

G Uniform avoidance of residue classes

Here is the finite predicate behind (S2). Use the core $C = \{3, 5, 7, 11\}$. For each $k = 1, \dots, 14$, distinguish whether 2 is present, and enumerate $S = C \cap \operatorname{supp}(q)$, with $|S| + \epsilon \le k$, $\epsilon = \mathbf{1}_{2 \mid q}$. Put $L = H_k$ if $\epsilon = 0$, and $L = H_k/2$ otherwise. When 2 is present, the nonforbidden parity is a progression with difference 2; in its indices each odd prime still forbids one class. Let $q_S = \prod_{p \in S} p$ and

$$V_S(L) = \min_{0 \le a < q_S} \#\{a < n \le a + L : \gcd(n, q_S) = 1\}.$$

CRT translates arbitrary core forbidden classes into zero classes; the minimum covers every translation and starting point. Compute it over its complete period, retaining complete periods when $L > q_S$. The independent formula for every count is

$$\sum_{d \mid q_S} \mu_{\mathrm{ar}}(d) \left(\lfloor (a+L)/d \rfloor - \lfloor a/d \rfloor \right).$$

The largest core period is 1155.

Let $p_1 < p_2 < \cdots$ be the primes above 11 and $t = k - \epsilon - |S|$. External primes r_j obey $r_j \geq p_j$; each forbidden class removes at most $\lceil L/p_j \rceil$ positions. The complete finite predicate is

$$V_S(L) - \sum_{j=1}^{t} \lceil L/p_j \rceil \geq 1 \quad \text{for every one of the 399 admissible } (k, \epsilon, S). \tag{A4}$$

It holds for the lengths in (S2). If the support has fewer than k primes, add distinct primes above 11; avoiding the extended support implies avoidance for the original. The proof therefore covers arbitrarily large external primes and all choices of forbidden classes. Prime powers do not change these classes. The finite enumeration concerns the core period and the case inventory, not an alleged finite enumeration of all prime supports.

H Uniform lower profiles for the physical CDF

This appendix proves the infinite extension of the square and cross certificates. It applies to $w = 4, 5, 6$, with the condition $\mathrm{rad}(M) \notin \mathcal{E}$ for $w = 5$. The admissible S, external lower bounds ℓ_j, and omitted complete supports are exactly those of (S11). Let $P = \prod_{p \in \mathcal{P}} p = 9699690$, and define

$$U_S(n) = \sum_{h \mid P_S} \mu_{\mathrm{ar}}(h) \lfloor n/h \rfloor, \qquad \rho_S = \prod_{p \in S} (1 - 1/p), \qquad E_S = \begin{cases} 0 & S = \varnothing, \\ 2^{|S|-1} & S \neq \varnothing. \end{cases}$$

For integer $n \geq 0$, $|U_S(n) - \rho_S n| \leq E_S$. Only the integer version is used without an extra floor.

H.1 Square counts and analytic tails

For $R \geq 1$ set

$$T_S(R) = \sum_{k \mid P/P_S} \mu_{\mathrm{ar}}(k) \lfloor R/k \rfloor \sum_{h \mid P_S} \mu_{\mathrm{ar}}(h) \lfloor R/(kh) \rfloor .$$

It counts positive pairs $(a, d) \in [1, R]^2$ with d a core unit and with no common prime from $\mathcal{P}$. With sums over primes, define

$$\begin{aligned} D_{w,S}(R) = T_S(R) &- R \sum_{j=1}^{w-|S|} U_S(\lfloor R/\ell_j \rfloor) \\ &- \sum_{19 < p \leq 199} \lfloor R/p \rfloor \, U_S(\lfloor R/p \rfloor) - \sum_{199 < p \leq R} \lfloor R/p \rfloor^2 . \end{aligned} \tag{A5}$$

An external marked prime eliminates denominators $d = pe$, retaining $(e, P_S) = 1$, so its loss is at most $R U_S(\lfloor R/\ell_j \rfloor)$. A remaining nonprimitive pair has a common prime above 19; the last two sums cover it. The union bound can overcount losses, which is safe. The R pairs with $d = 1$ are always available. Thus the number $V_M(R)$ of primitive full-unit positive pairs is at least $v_{w,S}(R) = \max(R, D_{w,S}(R))$. Adding $(0, 1)$ and both signs gives $A_M(R) = 1 + 2V_M(R)$.

For two distinct primitive positive-denominator vectors defining the same state, the determinant is a multiple of M. In the square its absolute value is at most $2R^2$. Under

$2R^2 < M$, it vanishes, and primitiveness plus positive denominators forces equality. Conversely every height-R witness can be divided by its ambient gcd, which is a unit modulo M. Therefore $\mathbb{P}(R_M \le R) = A_M(R)/M$ in this strict range.

The infinite tail of (A5) has rational coefficients. Put

$$\alpha_S = \rho_S \prod_{p \in \mathcal{P} \setminus S} (1 - p^{-2}), \quad C_S = \sum_{\substack{k \mid P/P_S,\ h \mid P_S \\ \mu_{\mathrm{ar}}(k)\mu_{\mathrm{ar}}(h)=1}} (1/k + 1/(kh)).$$

The positive terms use $\lfloor x \rfloor \lfloor y \rfloor \ge xy - x - y$; the negative terms use the product of arguments as an upper bound. Hence $T_S(R) \ge \alpha_S R^2 - C_S R$. The numerical premise $\sum_p p^{-2} < 4523/10000$ is itself proved: sum all primes through 20000, rounding each reciprocal square up to 10^{-12}, and add the integer tail bound $\sum_{n>20000} n^{-2} \le 1/20000$. The resulting rational sum is below $4523/10000$. Let

$$b_0 = \sum_{19 < p \le 199} p^{-2}, \quad e_0 = \sum_{19 < p \le 199} p^{-1}, \quad \beta = 4523/10000 - \sum_{p \le 199} p^{-2}, \quad o = \sum_j \ell_j^{-1}.$$

Then, for every integer radius,

$$D_{w,S}(R) \ge \gamma R^2 - FR, \quad \gamma = \alpha_S - \rho_S(o + b_0) - \beta, \quad F = C_S + E_S(w - |S| + e_0). \tag{A6}$$

Choose $c_w = 11/50, 21/100, 37/200$ respectively for $w = 4, 5, 6$. The finite predicate checks $\gamma > c_w$ in every class, then sets $c = (\gamma + c_w)/2$ and $K = \max(4, \lceil F/(\gamma - c) \rceil)$. Equation (A6) proves $V_M(R) \ge cR^2$ for every $R \ge K$. This is a proof of the entire infinite tail, not an extrapolation of counts.

Use $a(R) = 1 + 2v_{w,S}(R)$ at all integers $1 \le R < K$, and $a(R) = 1 + 2cR^2$ thereafter. The ratio in the latter tail is increasing when $2cR \ge 1$, which is checked at K. Its value at K therefore bounds all later ratios. Define

$$\widetilde{h}(r) = \begin{cases} 0 & r = 0, \\ \min\left(\frac{1+2cK^2}{(K+1)^2}, \min_{r \le R < K} \frac{a(R)}{(R+1)^2}\right) & 1 \le r < K, \\ \frac{1+2cK^2}{(K+1)^2} & r \ge K, \end{cases} \qquad h_{w,S}^{1/2}(r) = 10^{-36} \left\lfloor 10^{36} \widetilde{h}(r) \right\rfloor. \tag{A7}$$

This is a minimum over all future radii, including the analytic tail. It is nondecreasing and gives $A_M(R)/(R+1)^2 \ge h(r)$ whenever $R \ge r$. The finite certificate must reconstruct every radius before K, not merely selected breakpoints.

H.2 Cross counts and analytic tails

For $A \in \mathbb{Z}_{\ge 0}$, put $b_A = \lfloor A/2 \rfloor$ and

$$f_S(A) = (A - b_A)U_S(b_A) + b_A U_S(A), \quad T_S^{\times}(R) = \sum_{n \mid P/P_S} \mu_{\mathrm{ar}}(n) f_S(\lfloor R/n \rfloor).$$

This is the cross count with core unit denominators and no common core prime. The two arms are counted with their intersection removed. Dividing by n preserves the shape because $\lfloor \lfloor R/2 \rfloor / n \rfloor = \lfloor \lfloor R/n \rfloor / 2 \rfloor$. For $r = \lfloor R/2 \rfloor$ define

$$\begin{aligned} D_{w,S}^{\times}(R) = T_S^{\times}(R) &- \sum_j [(R - r)U_S(\lfloor r/\ell_j \rfloor) + rU_S(\lfloor R/\ell_j \rfloor)] \\ &- \sum_{19 < p \le 199} f_S(\lfloor R/p \rfloor) - \sum_{199 < p \le R} f_{\varnothing}(\lfloor R/p \rfloor). \end{aligned} \tag{A8}$$

Thus $V_M^{\times}(R) \geq \max(R\mathbf{1}_{R\geq 2}, D_{w,S}^{\times}(R))$. At $R = 1$, the positive quadrant is empty, but $(0, 1)$ is included in $A_M^{\times}(R) = 1 + 2V_M^{\times}(R)$. Two cross vectors have determinant of absolute value at most $R^2 + r^2 \leq 5R^2/4$. Consequently $\mathbb{P}(R_M \leq R) \geq A_M^{\times}(R)/M$ if $5R^2 < 4M$. This is a lower bound, since the cross is only part of the physical ball. For example $M = 5, R = 2$ has a collision at the closed boundary, so the strict inequality is essential.

For every real $x \geq 0$, direct use of the integer unit count gives

$$\tfrac{3}{4}\rho_S x^2 - (\tfrac{9}{4}\rho_S + E_S)x \leq f_S(\lfloor x \rfloor) \leq \tfrac{3}{4}\rho_S x^2 + E_S x.$$

Indeed $3A^2/4 - 3A/4 \leq 2Ab_A - b_A^2 \leq 3A^2/4$, and $A^2 \geq x^2 - 2x$ for $A = \lfloor x \rfloor$. It follows that

$$D_{w,S}^{\times}(R) \geq \gamma^{\times} R^2 - F^{\times} R, \quad \gamma^{\times} = \tfrac{3}{4}\gamma,$$

$$F^{\times} = (\tfrac{9}{4}\rho_S + E_S) \sum_{\substack{n \mid P/P_S \\ \mu_{\mathrm{ar}}(n)=1}} n^{-1} + E_S \sum_{\substack{n \mid P/P_S \\ \mu_{\mathrm{ar}}(n)=-1}} n^{-1} + E_S(w - |S| + e_0). \tag{A9}$$

Use baseline $63/400$ for $w = 5$, $111/800$ for $w = 6$, average it with $\gamma^{\times}$, and choose K exactly as in (A6)–(A7). Checking $\gamma^{\times}$ exceeds the baseline proves the infinite tail. All finite counts before K, the suffix minima and their downward rounding then give $h_{w,S}^{4/5}$ by the same construction.

H.3 Uniform modulus comparison and integration

For $0 < u < \alpha$ let $R = \left\lfloor \sqrt{uM} \right\rfloor$. The strict injectivity condition holds, and $(R+1)^2/M > u$. If $R \geq 1$, monotonicity of h gives the first branch of (S6); if $R = 0$, its right side is zero. For $\alpha \leq u \leq 1$, use $R = r_\alpha(M)$, for which $R^2 < \alpha M \leq (R+1)^2$. Since $r_\alpha(M) \geq r_\alpha(Q)$, the second branch follows. This pays perfect-square endpoints without extending injectivity there.

The identity $L(M) = \frac{1}{2}\int_0^1 \mathbb{P}(Z_M \leq \sqrt{u})\, du/u$ proves (S7). Changes between $<$ and $\leq$ at isolated jump points do not affect the integral. For the square alone, writing $r = r_{1/2}(Q)$,

$$I_{4,S}(Q) = \frac{1}{2Q} \sum_{j=0}^{r-1} (2j+1) h_{4,S}^{1/2}(j) + h_{4,S}^{1/2}(r) \left(\frac{1}{4} - \frac{r^2}{2Q} + \frac{\log 2}{4} \right). \tag{A10}$$

For the joint profile all breakpoints $j^2/Q, 1/2, 4/5$, and any line/plateau intersections are rational. On a segment $[l, r]$ the maximum has form $mu + b$, $m, b \geq 0$, and its contribution is

$$\frac{m(r-l) + b\log(r/l)}{2}. \tag{A11}$$

When $l = 0$, $b = 0$ and the logarithm is omitted. This provides an exact finite integration rule for each fixed support.

I Uniform marked profiles and the excluded supports

This appendix gives the analytical contract of the finite profiles and the partitions used in the exclusions. Hermite normal form, Gauss reduction, the Chinese remainder theorem, Möbius inversion, Farey neighbours, and the classical Jacobsthal inequality are used in their usual senses. A finite profile is a collection of witnesses and outward bounds satisfying the inequalities below. A recorded successful run, without those inequalities and the stated coverage, is not a premise.

I.1 Signed transfer, including unequal valuations

For a squarefree mark $q \mid M$ define

$$R_{M,q}(s) = \min_{d>0,(d,q)=1,\ a\equiv sd \pmod M} \max(|a|, d), \qquad \nu_q(M) = \frac{1}{M}\sum_s \log R_{M,q}(s) - \frac{1}{2}\log M.$$

Thus $\mu(M) = \nu_{\operatorname{rad} M}(M)$. Intermediate marks are relaxations; they are not physical means until the full mark is restored. For $C \in \mathrm{GL}_2^+(\mathbb{R})$ and a column $m \bmod q$ nonzero modulo every prime of q, put

$$F_q(C, m) = \min_{n\in\mathbb{Z}^2,(nm,q)=1} \log \frac{\|nC\|_\infty}{\sqrt{\det C}}. \tag{E1}$$

An index-K sublattice has a unique representative $H = \left(\begin{smallmatrix} a & 0 \\ b & d \end{smallmatrix}\right)$, $ad = K, 0 \le b < a$. When $\operatorname{rad} K = q$, retain those matrices whose reduction has rank one at each prime of q. Their local kernels are the lines in $\mathbb{P}^1(\mathbb{F}_p)$. Each product of local kernel lines occurs K/q times: a line modulo p has p^{e-1} cyclic lifts at index p^e. For each parent mark exactly K branches satisfy $Hm \neq 0$ at every prime. Define $T_K f = K^{-1}\sum_{H \text{ admitted}} f(HC, Hm)$. The finite local transform is $\bigotimes_{p|q}(J_{p+1} - I_{p+1})$, with the multiplicity just stated. It acts on signed values. Replacing negative entries by zero inside this transform would change the proof.

On the physical matrix $C_{L,s} = \left(\begin{smallmatrix} L & 0 \\ s & 1 \end{smallmatrix}\right)$ with $m = (0, 1)^t$, admission forces $d = 1$. Its children are exactly $C_{KL,s+bL}$, $0 \le b < K$. Consequently

$$\nu_q(KL) = \frac{1}{L}\sum_{s=0}^{L-1}(T_K F_q)(C_{L,s}, m). \tag{E2}$$

Iterating the squarefree transfer gives the same identity for $K = q^k$. No coprimality of L with q is required. This proves applicability to unequal positive valuations and to additional primes in L.

I.2 Complete geometric profile certificates

A shortest Euclidean vector is primitive in its lattice and completes to a basis. Gauss reduction, scaling and rotation give

$$Y(\rho, t) = \begin{pmatrix} \rho & 0 \\ t\rho & 1 \end{pmatrix}, \qquad 0 < \rho \le 2/\sqrt{3}, \quad |t| \le 1/2. \tag{E3}$$

The rotation and the transformed mark remain in the state; the infinity norm does not permit discarding the rotation. On a cell in (ρ, t), an admitted integer witness n in each child has transport $(u, v) = nH$, and normalized vector

$$\frac{(\rho(u + tv), v)\operatorname{Rot}(\theta)}{\sqrt{K\rho}}. \tag{E4}$$

For fixed θ, each component is affine in t, and in $z = \frac{1}{2}\log\rho$ is $Ae^z + Be^{-z}$. Its absolute value is convex: on each smooth piece its second derivative equals its absolute value, and zeros have upward derivative jumps. Therefore the norm is bounded by its maximum at the four cell vertices. For each branch take the maximum over the four vertices before averaging the logarithmic costs over admitted branches; the order cannot in general be reversed. For a fixed vector, an active angular log-norm has second derivative at least -2, with upward derivative jumps. Nodes $\theta_i = \arctan(i/A)$, $-A \le i \le A$, cover a period; successive gaps are at most $1/A$. The value between nodes is at most the larger endpoint value plus $1/(4A^2)$. This angular

bound survives maxima and averages with nonnegative weights summing to one, including the branchwise vertex maxima.

These bounds have an exact finite checker. At a vertex $\rho = x/D, t = j/40, \theta = \arctan(n/A)$, the logarithm in (E4) is

$$\begin{aligned} &\log\max(|U|,|V|) \\ &\quad - \tfrac{1}{2}\{\log 1600 + \log D + \log(A^2+n^2) + \log K + \log x\}, \\ &U = Ax(40u+jv) + 40Dnv, \qquad V = -nx(40u+jv) + 40DAv. \end{aligned} \tag{E5}$$

Thus every branch witness, full-mark coprimality, vertex and angular node can be checked with integers and outward logarithms. At every angular node the cell cap must dominate the admitted-branch average of the individual vertex maxima, plus the interpolation cost. An aggregate cell maximum or a sampled audit alone does not establish this predicate.

Below the first radial endpoint ρ_0, CRT gives child coefficients in $[0, q-1]^2$. After transport their coefficients have absolute values at most $2(q-1)K$ and $(q-1)K$. Substitution in (E4) gives

$$T_K F_q \le \log(q-1) + \tfrac{1}{2}\log K - \tfrac{1}{2}\log\rho + \tfrac{5}{2}\rho. \tag{E6}$$

Take the maximum over marks, shears and angles, and the running maximum to the right of the cell caps. The resulting radial step heights $b_0 \ge \cdots \ge b_{J-1}$, upper endpoints r_j, and (E6) give a uniform mean bound

$$\begin{aligned} A(q,K) = b_{J-1} &+ \sum_{j=0}^{J-2} (b_j - b_{j+1}) \min(1, \alpha r_j) \\ &+ \alpha\rho_0 \left(\max(c - b_0, 0) + \tfrac{1}{2}\right), \\ c = \ &\log(q-1) + \tfrac{1}{2}\log K - \tfrac{1}{2}\log\rho_0 + \tfrac{5}{2}\rho_0. \end{aligned} \tag{E7}$$

Here $\alpha = 3/2$ by the primitive shortest-vector bound above; $\mathbb{E}\log(\rho_0/\rho)_+ \le \alpha\rho_0$ proves the cusp cost. Earlier profiles can use the weaker $\alpha = 29/14$: counting all integer incidences in the disk gives at most $\sum_{d=1}^{\lfloor Y\rfloor} 2\sqrt{Y^2-d^2} + \lfloor Y\rfloor(\lfloor Y\rfloor+1)/2 \le (\pi+1)Y^2/2 < 29Y^2/14$. The trapezoid estimate and $2\sqrt{z} \le z+1$, $z = Y^2 - \lfloor Y\rfloor^2$, prove the middle inequality. The larger cusp coefficient 4ρ is also permissible. Equations (E2)–(E7), rather than observations at finitely many moduli, prove $\nu_q(M) \le A(q,K)$ for every $K \mid M$.

Some iterated profiles use directional interpolation instead of retaining every descendant vector. The maintained inequality is

$$f(CA, m) \le b + \sum_j w_j \log\frac{\|u_j A\|_\infty}{\sqrt{\det A}} + e\,\mathrm{cond}(A)^2, \quad w_j \ge 0, \quad \sum_j w_j = 1. \tag{E8}$$

Directions are consecutive points on a square perimeter with mesh $1/T$. Linear interpolation costs at most $t(1-t)\mathrm{cond}(A)^2/T^2$. Rounding directional mass by total variation η costs at most $\eta\mathrm{cond}(A)^2$, since the oscillation is at most $\log(2\mathrm{cond}(A)) \le \mathrm{cond}(A)^2$. Transfer each admitted branch of (E8), retain all positive masses, and add every propagated error. The condition bound on a cell is checked at its vertices: the trace divided by the determinant has form $a(t)\rho + b/\rho$ with a convex and $b > 0$. If $k/10^9$ bounds the squared condition number, the exact test is

$$(k^2 + 10^{18})d^2 \ge k10^9(\mathrm{tr}^2 - 2d^2), \quad k \ge 10^9. \tag{E9}$$

Outside a directional grid, choose a shortest vector u in a child of determinant D, set $\nu = \|u\|_2^2$, and let K_0 be a coefficient-height bound. A valid fallback is

$$\begin{aligned} &\log(q-1) + \tfrac{k}{2}\log q + \tfrac{1}{2}\log D + \log K_0 - \log\nu \\ &\quad + \log\frac{\|(u^\perp/K_0)A\|_\infty}{\sqrt{\det A}} + \frac{4\nu}{D}\mathrm{cond}(A)^2. \end{aligned} \tag{E10}$$

Its coefficients follow from the same CRT completion as the cusp bound. Substitution checks the inequality before the logarithm. Equations (E8)–(E10) are part of the finite certificate, with every route and its rounding error retained. Finally (E7) integrates the profile, using its declared α and cusp coefficient.

The residual three-prime profiles also admit a Euclidean variant. Replace the infinity norm in (E1) by the Euclidean norm, which gives an upper bound for (E1). The squared transported norm is $\rho(u+tv)^2/K + v^2/(K\rho)$, separately convex in ρ, t. The four vertices suffice and rotation disappears. The signed transfer, mark tests, cusp and radial integration remain mandatory.

I.3 Restoring a prime: a uniform analytical lemma

Let $p \nmid q$, $p \mid M$, and choose a deterministic minimum $u = (a, d)$ marked q, of height H. If $p \mid d$, then $p \mid a$. The minimum is primitive in its lattice coordinates; dividing a lattice multiple would preserve the mark and lower its height. Complete it to a basis $u, v = (b, e)$ of determinant M, with $\gcd(d, e) = 1$, and write $v = \tau u + Mu^{\perp}/(a^2+d^2)$, $|\tau| \le 1/2$. With $\eta = ed^{-1} \bmod q$, find $1 \le i \le r < p$ and j such that $\gcd(j + i\eta, q) = 1$, $|j + i\tau| \le 1$. Then $ju + iv$ has a full-unit denominator modulo qp. For two primes in q, $r = 2$ works: try $(1, 0)$, $(1, -\varepsilon)$, then $(2, -\varepsilon)$, where $\varepsilon = 1$ for $\tau \ge 0$ and -1 otherwise; in the last case the two obstructions are at different primes. For three odd primes, $r = 3$ works: if η is not a unit, a prime dividing it never obstructs $i\eta - \varepsilon$, and each of the other two primes obstructs at most one of $i = 1, 2, 3$. For $q = 2uv$, u, v odd primes, $r = 4$ works. If η is even, the three odd numbers $\eta - 1, 2\eta - 1, 3\eta - 1$ are pairwise coprime. If η is an odd nonunit, one odd prime dividing it never obstructs $2\eta - 1, 4\eta - 1$, while the other cannot obstruct both. Reflect η, τ for a negative shear. A unit uses $(1, 0)$. These arguments supply the entire interval of shears, including zero.

The coarse bound is $R_{M,qp} \le H + rM/H$. A sharper bound, used for the later four-prime families, is

$$\log \frac{R_{M,qp}}{H} \le h_r(X), \quad X = H^2/M, \qquad h_r(X) = \begin{cases} \frac{1}{2}\log(1 + r^2/X^2), & X \le 3r/4, \\ \log(1 + r/(2X)), & X \ge 3r/4. \end{cases} \tag{E11}$$

Indeed, after expressing a repaired component in the basis $u, u^{\perp}$, its ratio to H is at most $\max(1 + v\xi/(1+\xi^2), \xi + v/(1+\xi^2))$, $0 \le \xi \le 1$, $v = r/X$. These expressions are bounded by $1 + v/2$ for $v \le 4/3$, and by $\sqrt{1+v^2}$ otherwise.

Write $u_q = \varphi(q)/q$, $U_q(A) = \#\{1 \le e \le A : (e, q) = 1\}$, $S_q(A) = \sum_{e \le A, (e,q)=1} e$. For $A = qn + j$, periodicity reduces

$$2AU_q(A) + S_q(A) \le a_q A^2 + c_q A, \qquad a_q = 5u_q/2 \tag{E12}$$

to the two coefficient inequalities $3\epsilon_j \le c_q$ and $2jU_q(j) + S_q(j) - a_q j^2 \le c_q j$, $\epsilon_j = U_q(j) - u_q j$, for every residue j. For a bad minimum, $a = pb, d = pe$. With $A = \left\lfloor \sqrt{MX}/p \right\rfloor$, $g = \gcd(e, M/p)$, the congruence has pg solutions if $g \mid b$, and zero otherwise. Hence the union of all bad incidences is at most $p\{2AU_q(A) + S_q(A)\}$. The numerator zero is retained. It follows for all real $x > 0$ that

$$\mathbb{P}(p \mid d_s, H_s^2/M \le x) \le \min(1, a_q x/p + c_q\sqrt{x/M}). \tag{E13}$$

Discarding the mark gives the weaker $\min(1, 3x/p)$. No distributional independence of the minima is assumed.

For $\Phi(y) = \log(1+y) + y\log(1+1/y)$, the coarse repair integrates to

$$0 \le \nu_{qp}(M) - \nu_q(M) \le \Phi(ra_q/p) + c_q\pi\sqrt{r}/\sqrt{M}. \tag{E14}$$

For (E11), when $ra_q/p \le 4/3$, the stronger alternatives are

$$E_{\mathrm{per}}(q,p,M) = \Phi(ra_q/(2p)) + \frac{ra_qK_*}{p} + \frac{c_qJ_r}{\sqrt{M}},$$
$$E_{\mathrm{free}}(p) = \Phi(3r/(2p)) + \frac{3rK_*}{p}, \tag{E15}$$
$$K_* = \arctan(3/4) - \tfrac{1}{2}\log(5/2) < 46339/250000, \quad J_r = \sqrt{r/4}\,J_4, \quad J_4 < 51/10.$$

These follow by integrating the nonnegative density against $-h_r'$: $\int_0^\infty \min(1,ax)(-h_r'(x))dx = \Phi(ra/2) + raK_*$. For the square-root error,

$$J_4 = 4\int_0^{\sqrt{3}/2} \frac{dt}{1+t^4} + 4\int_0^{2/\sqrt{3}} \frac{dt}{t^2+2}.$$

The alternating expansions through degrees 80 and 80, with the next-term bounds and rational upper endpoints $866026/10^6, 1154701/10^6$, prove the quoted rational bound. The arctangent alternating series and the logarithm series given below certify K_*. These are finite scalar inequalities extending the analytical integrals; they are not sampled repairs.

I.4 Four-prime families and eleven five-prime exceptions

For every row in the following table the prime parameter satisfies $p \nmid q$; all positive valuations and all lengths are allowed. The first gate covers $p \ge P$ at every length, the second covers $p \ge p_1, N \ge 2$, and the third covers $p \ge p_1, B \ge L$ at every length. Smaller primes are covered by full marked profiles at the declared indices and their finite physical complements. The last column counts the complete complement at $N = 1, p_1 \le p < P, B < L$; the two summands are the analytical rows and the physical witness rows.

q	$p_{\min}$	P	p_1	L	finite complement
30	11	20000	211	4000000	$24089 + 8811$
42	11	3400	107	4000000	$10028 + 2269$
70	11	1000	73	3000000	$2855 + 663$
66	13	350	47	1000000	$1118 + 357$
105	11	125	31	1000000	$270 + 117$
78	17	200	41	250000	$304 + 182$
110	13	160	37	200000	$185 + 107$
102	19	160	37	200000	$188 + 117$
154	13	120	37	200000	$102 + 46$
165	13	72	29	150000	$43 + 25$
130	17	125	37	180000	$105 + 59$
114	23	150	37	200000	$177 + 90$

(E16)

Here are the core profiles needed for those gates; $K : a$ means $K \mid M \Rightarrow \nu_q(M) < a$, proved by (E1)–(E10).

q	$K : a$
30	$30 : 7/25, \quad 900 : 61/500$
42	$42 : 107/500, \quad 1764 : 67/1000$
70	$70 : 7/50, \quad 4900 : 0$
66	$66 : 127/1000, \quad 4356 : 16/1000$
105	$105 : 6/125, \quad 11025 : -2/25$
78	$78 : 3/40, \quad 156 : 7/100, \quad 6084 : 1/250$
110	$110 : 17/500, \quad 220 : 23/1000, \quad 12100 : -6/125$
102	$102 : 6/125, \quad 204 : 9/200, \quad 10404 : -1/100$
154	$154 : 1/500, \quad 308 : -7/500, \quad 1078 : -49/1000, \quad 1694 : -61/1000$
165	$165 : -21/500, \quad 495 : -31/500, \quad 825 : -19/250, \quad 1815 : -2/25$
130	$130 : 7/500, \quad 260 : 3/1000, \quad 650 : -7/400, \quad 1690 : -21/500$
114	$114 : 1/25, \quad 228 : 31/1000, \quad 342 : 3/100, \quad 2166 : -1/200$

(E17)

For $N \geq 2$, divisibility by an index such as 825 or 1690 follows from the two positive valuations at its squared prime; it does not require equal valuations. At $N = 1$ additional mixed indices are used only when they actually divide $M = B$. Every gate substitutes its smallest permitted p, M, B into $(a + E(q, p, M))/\log B$. The numerator bounds decrease in p, M and $\log B$ increases. For the early 30 and 42 gates use (E14); for the other gates use (E15), with (E12) checked for the actual core. The particularly useful explicit early cost is

$$E(30, p, M) = \Phi(8/(3p)) + 7304/(399\sqrt{M}), \quad a_{30} = 2/3, \quad c_{30} = 166/57. \tag{E18}$$

The periodic coefficients in (E12), the gate comparisons, and all small-prime marked profiles form finite parts of the certificate. Their full marks are indispensable.

For clarity the small-prime intervals in (E16) are respectively $11 \leq p \leq 31$ and $37 \leq p < 211$ for core 30, $11 \leq p \leq 103$ for 42, $11 \leq p \leq 71$ for 70, $13 \leq p \leq 43$ for 66, $11 \leq p \leq 29$ for 105, $17 \leq p \leq 37$ for 78, $13 \leq p \leq 31$ for 110, $19 \leq p \leq 31$ for 102, $13 \leq p \leq 31$ for 154, $13 \leq p \leq 23$ for 165, $17 \leq p \leq 31$ for 130, and $23 \leq p \leq 31$ for 114. These are intervals of primes, not of all integers. Profile bounds at $K = qp$, or at its declared repeated or mixed index, cover the tails in B, N; every smaller positive power product is in the finite witness complement. In the 30 row the seven early primes have complete correction bound 233/10000, and each of the 35 primes $37 \leq p < 211$ has $\mu(M) < 11/100$ for every valuation. The special prime $p = 7$ is the support 210, treated separately below.

The eleven five-prime radicals in the main text are handled as follows. For 2310, reintegrating its complete radial profile with $\alpha = 3/2$ gives a correction bound below 0.022928430661275. The $K = q$ profile covers $N = 1, B \geq 100000$; the remaining 30 bases are physically certified, and $K = 6q$ covers every $N \geq 2$. For 2730, the $K = q$ profile covers $N = 1, B \geq 20000$, the seven bases $q, 2q, \ldots, 7q$ are physically certified, and $K = 6q$ covers every $N \geq 2$, giving a correction bound below 0.023441723318271. For $3570, 3990$, use the $K = 2q$ profile whenever $Nv_2(B) \geq 2$. Otherwise $N = 1, v_2(B) = 1$: the base q is physically certified, and every larger such base is at least $3q$, where the $K = q$ profile applies. The correction bounds are below $0.022646842649968, 0.022785731780385$, respectively. For each of $4290, 4830, 5610, 6006, 6090, 6270, 6510$, a single $K = q$ profile satisfies $\max(0, A(q, q))/\log q < 2/85$, so covers all positive valuations and lengths directly. This includes the mark 114 insertion family and its full five-prime exception 6510, without identifying their two different observables.

I.5 Finite physical witnesses and the support 210

For each finite modulus list use all representatives $0 \le s \le \lfloor M/2 \rfloor$. Verify

$$d_s > 0, \quad \gcd(d_s, M) = 1, \quad a_s \equiv s d_s \pmod{M}. \tag{E19}$$

Assign weight one to zero and to $M/2$ when M is even, and two otherwise. These weights sum exactly to M, by the admissible symmetry $(a, d) \mapsto (-a, d)$. For $H_s = \max(|a_s|, d_s)$,

$$\mu(M) \le \frac{1}{M} \sum_s w_s \log H_s - \tfrac{1}{2} \log M. \tag{E20}$$

The domain is enumerated both by positive prime powers and independently by factorization of the bounded integers. All records, order, counts, weights and the exact end of the witness stream are checked. Minimality is unnecessary for an upper bound. A histogram without verified admissible witnesses is insufficient.

For radical 210, the profile bounds are $\mu(M) < 7/25$ if $210 \mid M$, $\mu(M) < 27/200$ if $210^2 \mid M$, and $\mu(M) < 61/500$ if $210^3 \mid M$. The first covers $B \ge 150000$ at every length. Below that threshold, $N = 1$ has 121 positive-power bases, including 210. For $B \ne 210, N \ge 2$, $B \ge 420$ and the second profile suffices. For $B = 210, N \ge 3$, the third suffices. The remaining square $B = 210, N = 2$ has physical upper bound $14135527587997507 27/22050000000000000000$. Only the candidate $(210, 1)$ needs exact minima: its product is

$$P = 2^{76}3^{57}5^{25}7^{21}11^{24}13^{26}17^{20}19^{26}23^{20}29^{4}31^{4}37^{8}41^{2}53^{2}, \quad c_0 = \frac{\log P}{210 \log 210} - \frac{1}{2}. \tag{E21}$$

The denominator-one bound $R \le 105$ reduces exact minimality to the finite full-unit denominators up to 105; all 210 states are checked. The integer inequality $P^{170} > 210^{18690}$ proves $c_0 > 2/85$. Its outward enclosure is the one in the main text. Together the profiles and physical complement show that the entire radical-210 sector has correction at most c_0. This sector statement does not assert attainment of the global supremum.

I.6 The complete three-prime exclusion

The prime-power and two-prime theorems already proved in this paper give nonpositive correction in those sectors. For three primes, the following analytical formulas and finite complements give correction at most 2/85. We spell out the formulas to distinguish complete means from partial tail estimates.

First consider $\operatorname{rad} M = 6p$. For the primes $5 \le p \le 113$, 33 complete marked profiles give the following mean bounds, with index $(6p)^k$ in the column labelled k:

p	$k=1$	$k=2$	$k=3$	length-one cutoff	other finite cases
5	285/1000	1/8	31/500	$B \ge 200000$	$N = 2, B < 210$
7	107/500	67/1000		$B \ge 10000$	
11	129/1000	19/1000		$B \ge 250$	
13	21/200	7/1000		$B \ge 90$	
$17 \le p \le 113$	7/100			$B \ge 6p$	

(E22)

Each prime in the last line is checked separately. The finite complement consists of 157,43,3,1 length-one bases in the first four lines, and the six length-two bases $30, 60, 90, 120, 150, 180$. These are 210 base–length pairs and 204 different moduli. Applying (E19)–(E20) with the actual $\log B$ gives a common correction bound below $117/5000 < 2/85$. Every length and positive valuation is covered by either a row of (E22) or that complement.

For $p \ge 127$, a direct complete-mean formula is

$$\begin{aligned}\mu(M) &< \Phi(5/(3p)) - 1/200 + 17/(30p) \\ &\quad + \frac{20 + (31/20)\log A}{A} + \frac{13/4 + (4/9)\log A}{A^2}, \qquad A = \sqrt{M}.\end{aligned} \tag{E23}$$

Here is an explicit derivation contract. Let $P_m(T)$ count positive primitive $a, d \le T$ with $(d, m) = 1$, and put $\alpha_0 = 1/\zeta(2), \ell = \log 2, \beta = 3 - 2\sqrt{2}$. Möbius inversion and the period-six unit count give

$$|P_6(T) - \alpha_0 T^2/2| \le 4T/3 + (T/3)\log T, \quad P_{6p}(T) \ge P_6(T) - T^2/(3p) - 2T/3. \tag{E24}$$

For $T^2 \le M$, signed primitive witnesses give $C_6(T) \ge 1 + 2P_{6p}(T) - J(T)$, where $J = 0$ if $2T^2 \le M$ and otherwise $J \le L^2/18 + 8L/9 + 16/9$, $L = 2T - M/T$. This follows from the determinant argument and the interval bound $h/3 + 4/3$ for unit denominators in the triangular deficit region. For $T^2 > 2M$, the Farey bound (E27) below applies with $q = 6$. Its cancelled-denominator sum satisfies

$$\left| \sum_{d \le y, (d,6) > 1} \varphi(d)/d - \alpha_0 y/2 \right| \le 1 + (2/9)\log y \quad (y \ge 1).$$

Indeed, its exact Möbius expansion has bracket $\frac{1}{2}\lfloor y/(2g)\rfloor + \frac{2}{3}\lfloor y/(3g)\rfloor - \frac{5}{6}\lfloor y/(6g)\rfloor$, whose discrepancy from $y/(3g)$ lies in $[-2/3, 1/3]$ over every real period. The period-six harmonic and reciprocal-square tails give the stated error. Integrate the CDF below A, its plateau from A to $\sqrt{2}A$, and the survival above $\sqrt{2}A$, retaining the initial discrete interval. Add (E14) with $r = 2, a_6 = 5/6, c_6 = 13/6$. The negative constant before the costs of the third prime is $\alpha_0(11/16 + \ell/2) - 1/3 - 5\ell/12 > 1/200$. The coefficients of $A^{-1}, A^{-1}\log A, A^{-2}$ are respectively less than $20, 31/20, 13/4$; the last logarithmic coefficient is $4/9$, and the extra central loss is $(1 + \ell)/(3p) < 17/(30p)$. This proves (E23), with none of the third-prime central loss discarded. For $M \ge 150000$, evaluate the decreasing bound at $p = 127, A = 387$ to obtain $2085787481/14265497250 < 3/20$. Below 150000, all 8816 positive-power moduli are enumerated; 6397 pass (E23) and 2419 pass (E20) with bound $3/20$. Thus every $p \ge 127$ has $\mu(M) < 3/20$, giving correction $< 3/(20\log 762) < 2/85$. There is no prime between 113 and 127.

For the other triples write $r < s < p$, $q = rs \ne 6$, $Q = qp$. For each mark $m \in \{q, Q\}$ define

$$\begin{gathered}u_m = \varphi(m)/m, \quad \rho_m = \prod_{l \mid m} l/(l+1), \quad \epsilon_j = U_m(j) - u_m j, \quad 0 \le j < m, \\ D_m^+ = \max_j \epsilon_j, \quad D_m^- = u_m - \min_j \epsilon_j, \quad I_m = D_m^+ + D_m^-, \\ K_m = \max(D_m^+, 2u_m - \min_j \epsilon_j, u_m).\end{gathered}$$

Choose h_m to dominate both $\max_{1 \le n \le 16}(\sum_{j \le n, (j,m)=1} 1/j - u_m \log n)$ and $\sum_{j \le 16, (j,m)=1} 1/j - u_m \log 16 + (16u_m - U_m(16) + D_m^+)/16$. Partial summation proves $\sum_{j \le T, (j,m)=1} 1/j \le h_m + u_m \log T$ for every real $T \ge 1$. Periodic endpoint bounds give $|\,\lfloor x\rfloor\, U_m(\lfloor x\rfloor) - u_m x^2| \le K_m x$. Möbius inversion and the absolutely convergent Euler product then give

$$|P_m(T) - \alpha_0 \rho_m T^2| \le (K_m h_m + u_m)T + K_m u_m T \log T. \tag{E25}$$

Indeed the exact sum is $\sum_{g \le T, (g,m)=1} \mu_{\mathrm{ar}}(g)\, \lfloor T/g\rfloor\, U_m(\lfloor T/g\rfloor)$; the harmonic bound pays the floor errors and the reciprocal-square tail costs $u_m T$. The leading coefficient is $u_m \sum_{(g,m)=1} \mu_{\mathrm{ar}}(g)/g^2 = \alpha_0 \rho_m$.

For $T^2 \le M$, complete witnesses give

$$C_q(T) \ge C_Q(T) \ge 1 + 2P_Q(T) - J_M(T), \quad J_M(T) \le u_Q^2 L^2/2 + 2u_Q I_Q L + I_Q^2 \tag{E26}$$

when $2T^2 > M$, and $J = 0$ otherwise. Same-sign determinants are strictly smaller than M. Opposite signs collide only when $af + ed = M$; for fixed unit denominators d, f, the numerator window has length strictly less than one progression step because $T(d+f) - M - df = T^2 - M - (T-d)(T-f) < 0$. At $T^2 = M$, unit denominators are smaller than T, so the strict inequality still holds. The deficits lie in $x + y \le L$. Applying the interval bound $u_Q h + I_Q$ twice gives (E26). One may alternatively use $P_Q(T) \ge P_q(T) - u_q T^2/p - D_q^+ T$ and the weaker collision bound with u_q, I_q; this will be used for large p.

For the relaxed two-prime mark and $T^2 > 2M$,

$$M - C_q(T) \le \sum_{d<M/T,(d,q)>1} \varphi(d) \left(1 + \frac{2(M - Td)}{d(T-2d)}\right). \tag{E27}$$

To prove this, consecutive Farey fractions have determinant one and denominator sum greater than T. These properties follow by mediant insertion; a fraction between a determinant-one pair has denominator at least their sum. A unit centre a/d covers $|x - a/d| \le T/(Md)$. If both adjacent denominators are nonunit, their coprimality makes their two prime obstructions complementary; their difference is a unit. Thus every uncovered state is next to a cancelled centre with $d < M/T < T/2$. The neighbour, or its difference with that centre, has unit denominator greater than $T - 2d$. The resulting open hole has total length at most $2(M - Td)/(d(T-2d))$ in state coordinates. Each hole contains at most its length plus one integers, and there are $\varphi(d)$ centres of denominator d. This proves (E27), including the indispensable integer atom. This Farey argument is restricted to the two-prime mark, not extended to an arbitrary mark by analogy.

Set $a_r = 1 - 1/r, a_s = 1 - 1/s, a_q = 1 - 1/q$, $c'_q = a_r/r + a_s/s - a_q/q$, and

$$W(y) = a_r \lfloor y/r \rfloor + a_s \lfloor y/s \rfloor - a_q \lfloor y/q \rfloor - c'_q y,$$
$$D_W = \max_{0 \le j < q} (|W(j)|, |W(j) - c'_q|), \quad e_0 = D_W h_q + 2c'_q, \quad e_1 = D_W u_q, \quad \theta = \alpha_0(1 - \rho_q).$$

Expanding $\varphi(d)/d$ by Möbius gives the exact bracket $a_r \lfloor y/(rg) \rfloor + a_s \lfloor y/(sg) \rfloor - a_q \lfloor y/(qg) \rfloor$ in $\sum_{(g,q)=1} \mu_{\mathrm{ar}}(g)/g$. The periodic bound, harmonic estimate and $y \sum_{g>y} g^{-2} < 2$ prove cancelled density error $E(y) = e_0 + e_1 \log y$ for $y \ge 1$, and $E = e_0$ below one. Partial summation in (E27) gives, $t = T/\sqrt{M} > \sqrt{2}$,

$$\frac{M - C_q(T)}{M} \le V(t) + 4E(M/T)/T, \quad V(t) = \theta \sum_{j \ge 1} \frac{(2/t^2)^j}{j(j+1)} + \frac{\theta}{2t^2}.$$

The positive series has $V(\sqrt{2}) = 5\theta/4$ and $\int_{\sqrt{2}}^{\infty} V(t)dt/t = (1 - \rho_q)/2 - 3\theta/8$, by $\sum_{j \ge 1} 1/[j^2(j+1)] = \zeta(2) - 1$.

Let b_q be the least nonnegative coefficient certified by (E12). Define, with $u = u_Q, I = I_Q, c = 2\alpha_0\rho_Q$, $E_0 = 2(K_Q h_Q + u), E_1 = 2K_Q u$,

$$\Lambda_{\mathrm{full}} = \ell/2 + (1 - \rho_q)/2 - 3\theta/8 - \alpha_0\rho_Q(1 + \ell) + 3(1 - \ell)u_Q^2/4 + \Phi(5u_q/p),$$
$$A_0 = c(1 + \ell) + E_0(1 + \ell/2) + 2uI\beta + \ell uI + (1 + 5\theta/4 + 4e_0)/\sqrt{2} + b_q\pi\sqrt{2},$$
$$A_1 = E_1(1 + \ell/2) + 4e_1/\sqrt{2}, \quad A_2 = I^2\ell + 2e_0, \quad A_3 = 2e_1.$$

The complete-mean bound is

$$\mu(M) \le \Lambda_{\mathrm{full}} + (A_0 + A_1 \log A)/A + (A_2 + A_3 \log A)/A^2. \tag{E28}$$

For verification of the integration, write $\nu_q = -D + P_{\mathrm{mid}} + P_{\mathrm{far}}$. Equations (E25)–(E26) give

$$\begin{aligned} D &\ge c/2 - u^2(3/4 - \ell) - (c + E_0 + E_1 \log A + 2uI\beta)/A - I^2\ell/(2A^2), \\ P_{\mathrm{mid}} &\le (1 - c + u^2/2)\ell/2 + \ell(2c + E_0 + E_1 \log A + 2uI)/(2A) + I^2\ell/(2A^2), \\ P_{\mathrm{far}} &\le (1 - \rho_q)/2 - 3\theta/8 + (1 + 5\theta/4 + 4E(A))/(\sqrt{2}A) + 2E(A)/A^2. \end{aligned}$$

Use $\int_1^A \lfloor x \rfloor^2 \, dx/x \ge A^2/2 - A$, proved by summing $n^2 \log(1 + 1/n) \ge n - 1/2$; the last partial interval has remaining margin $1/2 + t - 3t^2/2 \ge 0$, $0 \le t \le 1$. Collision integrals are $A^2(3/2 - 2\ell)$ and $A\beta$. The far sum starts at $\left\lfloor \sqrt{2M} \right\rfloor + 1$; the first interval costs $1/(\sqrt{2}A)$, and $\sum_{T \ge L} T^{-2} \le L^{-1} + L^{-2}$. Adding these three bounds and (E14) with $r = 2$ proves (E28).

Using the alternative centre after (E26) gives another bound of the same form, with

$$\begin{aligned} \Lambda_{r,s}(p) &= T_0(r, s) + \Phi(5u_q/p) + u_q(1 + \ell)/p, \\ T_0 &= \ell/2 + 1/2 - 3\alpha_0/8 - \rho_q\{\alpha_0(5/8 + \ell) + 1/2\} + 3(1 - \ell)u_q^2/4. \end{aligned}$$

Its error coefficients are the preceding A_i with u, I, c replaced by $u_q, I_q, 2\alpha_0\rho_q$, and $E_0 = 2(K_q h_q + u_q + D_q^+)$, $E_1 = 2K_q u_q$. For $p \ge 53$, $\Lambda_{r,s}(p) \le \Lambda_{2,5}(p)$. To check this analytical domination set $x = 1/r, y = 1/s$, so $0 \le x \le 1/2, 0 \le y \le 1/5, u_q \ge 2/5$. With $K_0 = \alpha_0(5/8 + \ell) + 1/2$, $C_0 = 3(1 - \ell)/4$, the normalized derivative of T_0 is at least $K_0/(1 + x)^2 - 2C_0(1 - x) \ge 4K_0/9 - C_0$. The other terms cost at most $(5/53)\log(55/2) + (1 + \ell)/53$, strictly smaller than that lower bound. The same argument applies to y. The universal periodic bounds $D^{\pm} \le 2, I \le 4, K \le 3, b_q \le 8, D_W \le 2, c_q' < 1/2$ therefore imply

$$\mu(M) < \Lambda_{2,5}(p) + (70 + 14 \log A)/A + (18 + 4 \log A)/A^2. \tag{E29}$$

For example $\epsilon_j = \{j/r\} + \{j/s\} - \{j/q\} \in (-1, 2)$ gives the discrepancy bounds, and partial summation gives $b_q \le 8$. All error functions decrease for $A \ge 1$. At $p = 53, A = 1000$, (E29) is below 0.275, whereas $(2/85)\log 10^6 > 0.325$. Thus radical at least 10^6 is covered. Below that radical threshold, 855 pairs (r, s) have possible third primes. Each pair has a finite scalar p_{cut} where the pair-specific bound is less than $(2/85)\log(rsp)$. Decreasing numerators and increasing logarithms cover every larger prime. The remaining 4193 supports have modulus cutoffs at most 1085440; below them, 785 moduli pass the analytical bound and 7590 pass (E20) with negative mean. This covers all $p \ge 53$, all valuations and all lengths, without dividing by $\log M$.

The remaining 442 triples have primes at most 47 and omit the pair 2, 3. Of these, 353 satisfy the finite rational test $\Lambda_{\mathrm{full}}^+ < (2/85)(\log Q)^-$. For each, start at Q and double a scalar modulus cutoff until (E28) passes the same inequality. Verify the outward coefficients $A_0 \ge A_1, 2A_2 \ge A_3$, which prove decreasing errors. The largest cutoff is 74490839040. All positive-power products below cutoff are enumerated by both nested powers and a priority queue. There are 4117 moduli: 2454 pass (E28) and 1663 pass (E20) with negative mean. For a modulus $M = \prod p_i^{e_i}$, write $g = \gcd_i e_i$, $b_*(M) = \prod p_i^{e_i/g}$. Every representation $M = B^N$ has $N \mid g$ and $B = b_*^{g/N}$, so $\log B \ge \log b_*$. The analytical rows are tested against $(2/85)\log b_*$, covering all their base–length representations. There are 4227 such representations.

For the remaining 89 supports, the Euclidean variant of (E1)–(E7) gives a first-index profile below $(2/85)\log q$ for 85 supports. The other four have the following two mean caps:

q	$A(q, q)^+$	$A(q, q^2)^+$	$N = 1$ cutoff	finite bases
70	0.218699709338489	0.088846297509591	12000	28
110	0.152189312537973	0.052036789658452	700	4
130	0.130742618687066	0.041769725542389	300	2
105	0.130906519634302	0.016180460800994	300	1

(E30)

The second profile covers every $N \geq 2$; the first covers the length-one tail. All 35 remaining bases have negative physical witness mean. The analytic and finite partitions just described, together with (E22)–(E23), exclude the entire three-prime sector.

I.7 Outward arithmetic and the final logical implication

All critical constants and comparisons can be reduced to rational inequalities. For a positive rational, binary reduction gives $1 \leq y < 2$, and $z = (y-1)/(y+1)$ satisfies $0 \leq z < 1/3$. Use

$$\log y = 2 \sum_{j=0}^{n-1} \frac{z^{2j+1}}{2j+1} + E_n, \qquad 0 \leq E_n \leq \frac{2z^{2n+1}}{(2n+1)(1-z^2)}. \tag{E31}$$

Integer floors and ceilings preserve the outward direction. Square roots are enclosed by integer-square tests. For $\zeta(2)$, finite reciprocal-square sums with integral/trapezoid tails give rational intervals, which are inverted positively. The inequality $\pi < 22/7$ follows by integrating $x^4(1-x)^4/(1+x^2) > 0$; polynomial division gives the integral $22/7 - \pi$. Uniform profile comparisons, full residue-period predicates, support and modulus inventories, finite witnesses, and the final 621 sector integrals are distinct finite obligations. Their retained certificate index identifies each mathematical input and its complete checking rule. A fast check of retained arithmetic does not by itself replay every historical witness producer or independently recheck every geometric cost; that distinction is part of the reproducibility scope.

After these finite predicates are satisfied, the preceding analytical lemmas imply all the infinite sector bounds. Each residual support uses its own square or joint square–cross lower CDF; profiles from different supports are never combined inside a maximum. The sector table in the main text and the classical tail then give $\Delta(B, N) \leq U_*$ for every $B \geq 2, N \geq 1$. Taking the supremum preserves this common non-strict rational bound, and the strict rational inequality $U_* < 0.241933621228105$ proves the advertised strict decimal bound for the same C^*.

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